

JUST AUTOS

JA Portal — Handover

How the portal is built, where it runs, every connection it has,
and the operating procedures for every module.

Compiled 20 August 2026 · justautos.app · Internal — not for distribution outside Just Autos

JA Portal — Full Handover Document

How the portal is built, where it runs, every connection it has (and how to set up or change them), and operating procedures for every module.

Compiled 20 August 2026 from the live codebase. Supersedes `03_SYSTEM_OVERVIEW.md` / `05_INTEGRATIONS.md` (May 2026) where they conflict.

1. What the portal is

An internal management platform for **Just Autos**, covering two MYOB business entities:

- **JAWS** — Just Autos Wholesale (distribution arm, holds stock, ~14 distributors across Australia)
- **VPS** — Vehicle Performance Solutions (the workshop entity; runs on **Mechanics Desk**, which remains the system of record — see §7.2)

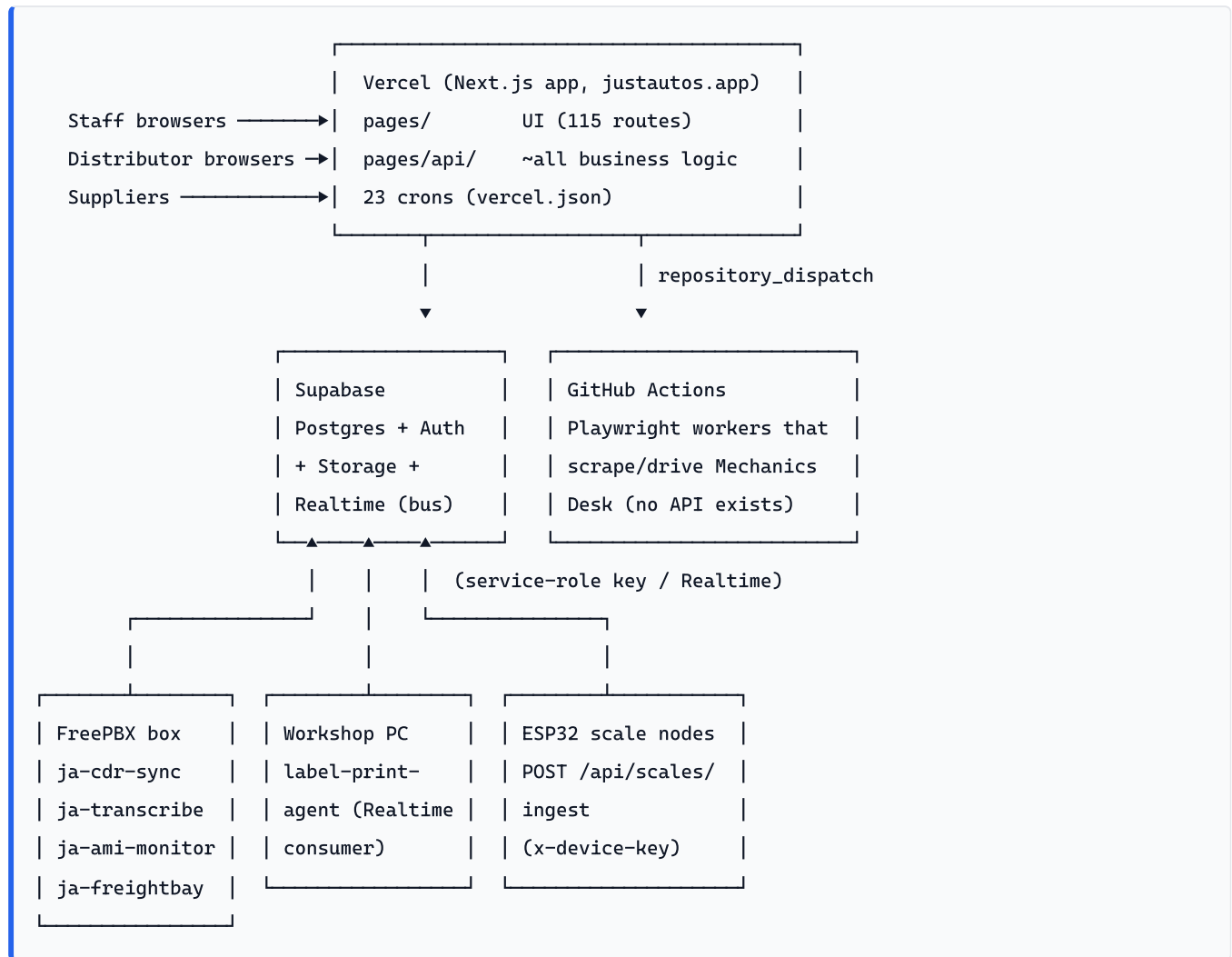
It is one Next.js application that contains: a **staff portal** (dashboards, workshop management, CRM, AP automation, calls coaching, reporting), a **distributor-facing B2B portal** (catalogue, checkout, orders, tune jobs, training) that went live in July 2026, a **supplier portal** (read-only stock wall), and a large fleet of **background automation** (25 Vercel crons, 16 GitHub Actions workflows, and several on-premise agents).

Production URL	<code>https://justautos.app</code>
Repo	<code>ChrisJustAutos/JA-Portal</code> , branch <code>main</code>
Hosting	Vercel (auto-deploys every push to <code>main</code>)
Database	Supabase project <code>qtiscbvhlvdvafwtdtcd</code> (<code>https://qtiscbvhlvdvafwtdtcd.supabase.co</code>)
Framework	Next.js 14.2.5, pages router , TypeScript, React 18
Key packages	<code>@supabase/supabase-js</code> , <code>@anthropic-ai/sdk</code> (mostly raw fetch is used), <code>exceljs</code> / <code>xlsx</code> , <code>@react-pdf/renderer</code> , <code>pdf-lib</code> , <code>leaflet</code> , <code>reactflow</code> , <code>sip.js</code> , <code>web-push</code> , <code>playwright</code> (dev, for GH Actions workers)

People

Person	Role
Chris Russell	Operations Manager — portal owner/direction
Nat	Accountant — chart of accounts, MYOB reconciliation sign-off
Matt Ashley	Technical — MYOB API, integrations
Matt H	Operations / Sales
Amanda	Accounts Payable — works in <code>/ap</code> daily
Laura	Director
Ryan	Receives the Monday 7am Weekly Sales Recap email

2. Architecture at a glance



External SaaS the Vercel app talks to directly: **MYOB AccountRight** (both entities, direct OAuth), **Xero** (migration target, foundation live), **Stripe** (B2B standalone account + JAWS read-only accounts), **Microsoft Graph** (mailbox webhooks + mail), **Resend** (email), **ClickSend** (SMS), **Slack** (bot + webhooks), **Monday.com**, **ActiveCampaign** (being replaced by CRM module), **MachShip** (freight), **Anthropic API** (18 LLM call sites), **Deepgram** (indirectly — runs on the PBX host), **Google Places** / **Nominatim** (geocoding).

Design conventions (must-follow)

- **API routes are .ts , never .tsx** — a .tsx in `pages/api/` crashes module load (generic 500 HTML).
- **Shared UI kit is mandatory:** `lib/ui/theme` tokens (`T` — CSS variables; use `alpha()` , never alpha-suffixed tokens) + `components/ui` + Feedback hooks. No `alert()` / `confirm()` browser dialogs. The B2B portal has its own separate "Alloy" kit at `components/b2b/ui.tsx` (one accent, $\geq 12\text{px}$ type, 44px touch targets).
- **Page scroll uses `minHeight: '100vh'` , never `height: '100vh'` + `overflow: 'hidden'`** — a clipped full-height wrapper wrapped around a `flex: 1` scroll pane leaves the bottom of the page unreachable (on a phone the wrapper is taller than the visible viewport and the document itself can't scroll). Staff `/admin/b2b/*` pages put their content in `<main className="b2b-admin-main">` , which also carries the shared mobile CSS that makes wide tables scroll sideways. This bit the Training page (Aug 2026) and Tune Jobs (fixed 2026-08-24).

- **Server Supabase clients** are constructed inline per route with the service-role key (module-level memo `_sb`). There is deliberately no shared server client module — follow the local pattern.
- **Long work (>~30s) never runs on Vercel** — anything browser-based or slow goes to GitHub Actions (dispatch pattern) or is chunked.
- **PostgREST silently caps every response at 1000 rows** (`db-max-rows`), whatever `.limit()` asks for — no error, no flag, the surplus is just dropped off the end of the sort order. Any select whose result set can exceed 1000 rows must page through `selectAllRows()` (`lib/supabase-paged.ts`), ordering on a **unique** column. This silently broke the Monday Quotes & Jobs Map report (see §7.8).
- **PostgREST embeds on workshop tables must use the `!customer_id` hint** (e.g. `workshop_customers!customer_id(...)`) or the query 500s on ambiguous relationships.
- **`b2b_distributor_users` is not unique per email/auth_user_id** (multi-site memberships) — never `maybeSingle()` on it.
- An **update notifier** compares the client bundle against `/api/version` and shows a "new version — Reload" banner after each deploy.

3. Environments, deployment & dev workflow

Deploy

1. Commit to `main` and **push** (convention: always push immediately after committing — never leave local-only commits).
2. Vercel builds and deploys production automatically. There is **no staging branch**; PR preview URLs exist if wanted.
3. Function logs: Vercel dashboard → Deployments → deploy → Functions tab.

Error fingerprints: generic 500 HTML page = module-load crash (bad import / `.tsx` in api / syntax error); JSON error = handler crash; empty response/timeout = function `maxDuration` exceeded (per-route overrides live in `vercel.json` → `functions`).

Database migrations (SOP)

- Migrations live in `migrations/NNN_description.sql` (202 files, `002` – `202`; note `148` and `153` are each duplicated — sequence is a convention, not a key. Next number: **203**).
- There is **no migration runner**. The procedure is: write the SQL file in `migrations/`, apply it to the live DB via the **Supabase MCP `apply_migration`** tool (project `qtiscbvhlvdvafwtdtcd`) **before** pushing code that depends on it, then commit both.
- The repo file is the source-of-truth record; the MCP apply is what actually changes the DB.

Local dev

`npm run dev` with a `.env.local` carrying at minimum the Supabase URL/keys. Most features hit live external services, so local dev is mainly for UI work; be careful with anything that writes to MYOB/Stripe.

Remote/mobile ops

A workshop MSI laptop runs OpenSSH + Tailscale + Claude Code; Chris's phone (Tailscale + Termius) can SSH in from anywhere. This is also the **only interactive route to the FreePBX box** (Tailscale IP `100.82.97.46`).

4. Auth & access model

Four separate authentication schemes coexist. They never overlap.

4.1 Staff portal

- Supabase Auth session, cookie `ja-portal-access-token`; identity in `user_profiles`. Opt-in TOTP 2FA (server-side enforced).
- **Six roles** (`lib/permissions.ts` is the source of truth): `admin`, `manager`, `sales`, `workshop`, `accountant`, `viewer`. (`lib/auth.ts` has a stale 5-role type missing `workshop` — known drift.)
- Roles map to `view:*` / `edit:*` / `admin:*` permission strings via `ROLE_PERMISSIONS`. Pages gate with `requirePageAuth(ctx, 'permission')`; API routes with `withAuth(permission, handler) / requireAdmin`.
- Two per-user allowlists narrow further: `user_profiles.visible_tabs` (which nav tabs/apps a user sees) and `visible_report_tabs` (which Reports sub-tabs).
- **User management:** `/settings?tab=users` — invite (Supabase invite email → `/reset-password?welcome=1`), change role, deactivate, delete. Audit log at `/settings?tab=audit`.
- Session persistence: `SessionKeeper` silently re-syncs cookies from `localStorage` (never on auth pages) and login pages silently resume valid sessions; deep links use `?next=`.

4.2 Distributor (B2B) portal

- Separate cookies `ja-b2b-access-token` / `ja-b2b-refresh-token`; identity in `b2b_distributor_users`. Gate: `requireB2BPageAuth`.
- Login = email+password with optional TOTP; also magic-link and signed invite tokens (`/b2b/welcome?t=...` — scanner-proof: opening does nothing, only the human submitting the set-password form activates).
- Users are `owner` or `member` per distributor; one auth user can belong to **multiple distributors** (account switcher, `ja-b2b-dist` cookie).
- Admin **preview mode**: signed `b2b_preview` token from the admin side renders the portal as a distributor with all non-GET requests blocked.
- `checkoutEnabled` per distributor = browse-only kill switch.

4.3 Supplier portal

Separate again: `b2b_supplier_users`, `requireSupplierPageAuth`, single read-only page (`/b2b/supplier`).

4.4 Machine auth

- **Service tokens** (`lib/service-auth.ts`): SHA-256 hashed rows in `service_tokens`, presented as `X-Service-Token` header (deliberately not `Authorization` to avoid clashing with Supabase JWTs). Scoped (`stocktake:write`, `calls:monitor`, `ap:admin`, `reports:read`, `upload:job-report`). Used by GitHub Actions workers and the PBX AML agent. Managed via `/api/admin/service-tokens`.
- **Signed capability tokens** in URLs for login-less pages: `/tune-jobs?token=` (distributor-scoped weekly reminder), `/order-action` (admin Book Freight email button).
- **Device keys**: ESP32 scale modules authenticate to `/api/scales/ingest` with `x-device-key` matched to `scale_devices`.
- **MCP tokens**: `jap_...` personal tokens (hashed in `mcp_tokens`) or OAuth 2.1 for the read-only Claude connector at `/api/mcp`.

- **Cron auth:** Authorization: Bearer \$CRON_SECRET (some handlers also accept the vercel-cron user-agent, and some accept a logged-in staffer with the right permission for manual runs).

5. Connections & integrations

5.1 Where credentials live (read this first)

lib/integration-config.ts implements a **DB-first resolver**: `getIntegration(key)` = `integration_settings` DB row → env var of same name → `''`. Values cache in-process for 30s. **Always use this resolver, not `process.env`, for the managed keys.**

Integration	Stored where	Change without redeploy?
ClickSend, Resend, CRM intake token, Xero app creds , accounting provider switch	<code>integration_settings</code> DB (env fallback)	Yes — Settings → Connections → Integrations; live in ~30s
MYOB app creds, Graph, Monday, ActiveCampaign, Stripe, Slack, Anthropic, VAPID, Supabase, GitHub dispatch	Vercel env vars only	No — edit env + redeploy
MYOB / Xero OAuth tokens	<code>myob_connections</code> / <code>xero_connections</code> tables	Yes — re-run connect flow
MachShip token	<code>b2b-freight-carrier-connections.credentials</code> JSONB	Yes — Admin B2B → Settings
Service tokens / MCP tokens / scale device keys	<code>service_tokens</code> / <code>mcp_tokens</code> / <code>scale_devices</code>	Yes — respective admin UIs

Admin surfaces:

- **Settings** → **Connections** (`/settings?tab=connections`) — three sub-tabs: **Integrations** (edit DB-managed credentials, test SMS/email, rotate CRM intake token, Xero connect), **Health** (live status board), **MYOB Connection** (connect/select company files).
- `/admin/connections` — standalone auto-refreshing health page (⚠ no SSR auth gate — relies on API-level gating).

5.2 MYOB AccountRight (primary accounting — both entities)

- **Code:** `lib/myob.ts` (OAuth + `myobFetch`), `lib/myob-reporting.ts` (replaced CData 2026-07-14 — reporting **must fetch all 4 invoice types**), plus per-module writers (`ap-myob-bill`, `b2b-myob-invoice`, `workshop-myob-invoice`, `stripe-myob-sync`, ...).
- **App creds (env only):** `MYOB_CLIENT_ID`, `MYOB_CLIENT_SECRET`, `MYOB_REDIRECT_URI`, `MYOB_SCOPE`.
- **Tokens:** `myob_connections` table, one row per label `JAWS` / `VPS`. Access ~20 min (auto-refresh), refresh ~1 year but **must be exercised regularly** — a multi-week quiet spell can kill the connection.
- **Connect / re-auth SOP:** Settings → Connections → MYOB Connection → Connect (or `GET /api/myob/auth/connect?label=JAWS|VPS`) → consent → pick company file → if the file is in legacy auth mode, set company-file username/password. Test with `/api/myob/test/invoice?label=VPS`.
- **Gotchas** (hard-won, in code comments):

- Two company-file auth modes — SSO (bearer only) vs legacy (`x-myobapi-cftoken`); header only sent when `company_file_username` is set.
- **Never page with bare \$top / \$skip** — no stable ordering, rows get skipped. Follow `NextPageLink` , page size 400.
- Every call is logged to `myob_api_log` ; the insert is `await` ed in `finally` (fire-and-forget logs were killed on timeout).
- **IsTaxInclusive matters for cent-exact invoices** — B2B invoices are pushed inc-GST with checkout-exact line pricing.
- No P&L endpoint exists in AccountRight (the old CData P&L panels were retired).

5.3 Xero (migration target — foundation live 2026-08-05)

- **Code:** `lib/xero.ts` , `lib/accounting/xero-adapter.ts` ; provider switch in `lib/accounting-provider.ts` .
- **App creds (DB-managed):** `XERO_CLIENT_ID` , `XERO_CLIENT_SECRET` , `XERO_REDIRECT_URI` — editable in the Integrations tab.
- **Setup SOP:** create a Web App at `developer.xero.com` with the redirect URI shown on the card → paste ID/secret → Save → "Connect Xero" (`/api/xero/auth/start`) → one consent covers both orgs → map tenants to `VPS / JAWS` via `/api/xero/connections` PATCH → ping to verify.
- **Provider switch:** `accountingProvider(entity, module?)` resolves `ACCOUNTING_PROVIDER_{ENTITY}_{MODULE}` → `ACCOUNTING_PROVIDER_{ENTITY}` → default `'myob'` . Entity-wide keys are in the Integrations UI; per-module overrides (AP, STATEMENTS, LETTERS, WORKSHOP, INVENTORY, B2B, REPORTING, STRIPE, BANK) are env-only. After cutover MYOB stays connected read-only for history.
- **Critical gotcha: Xero refresh tokens are single-use and rotate on every refresh** — the new token must be persisted or the connection dies. Refresh is serialised in-process; a cross-instance race shows as `invalid_grant` and self-heals next run. Granular scopes only (bills live under `accounting.invoices`); every call needs the `Xero-tenant-id` header.

5.4 Stripe

- **B2B checkout (standalone account `acct_1TrRkZ...` , LIVE keys):** `lib/stripe.ts` (raw fetch, no SDK). Env: `STRIPE_SECRET_KEY` , `STRIPE_WEBHOOK_SECRET` . Webhook POST `/api/b2b/stripe/webhook` (event `checkout.session.completed` ; `bodyParser: false` is required for signature verification — do not remove). Idempotent; if the MYOB writeback fails it still 200s and parks the error on `b2b_orders.myob_write_error` for manual retry from the admin order page. Payment methods live: **card (with surcharge), PayTo, BECS direct debit** (BECS settles later — a settle gate + `/api/cron/b2b-payment-check` confirm before fulfilment milestones).
- **JAWS reconciliation accounts (read-only):** `lib/stripe-multi.ts` , env `STRIPE_SECRET_KEY_JAWS_JMACX` , `STRIPE_SECRET_KEY_JAWS_ET` , labels registered in `STRIPE_ACCOUNT_LABELS` . Used by `/stripe-myob` push tool and payout reconciliation.

5.5 Microsoft Graph (mail)

- **Auth:** app-only client credentials. Env: `GRAPH_TENANT_ID` , `GRAPH_CLIENT_ID` , `GRAPH_CLIENT_SECRET` (+ webhook URLs and ~15 mailbox-assignment vars — see `lib/microsoft-graph.ts` and §22 of the integrations inventory). Azure app needs **Mail.Read, Mail.ReadWrite, Mail.Send with admin consent**

(Mail.Send 403 until granted — this bit us).

- **Webhooks:** `/api/webhooks/graph-mail` (Pipeline A — quote PDFs from rep mailboxes), `/api/webhooks/graph-jobreport-mail` (nightly MD WIP report).
- **Subscription lifecycle SOP:** subscriptions max out at ~70.5h. The cron `/api/cron/renew-graph-subscriptions` (every 6h) extends anything within 24h of expiry. **Recreating a dead/failed subscription is NOT automated** — re-run `POST /api/admin/setup-graph-subscriptions?key=$GRAPH_ADMIN_SETUP_SECRET` (idempotent).
- **Adding a rep mailbox** touches three places: `lib/agents.ts` (owner map), `MAILBOX_FOR_NAME` in `pages/api/cron/health-check.ts`, and re-running the setup endpoint.
- Inbox-driven pipelines that *poll* Graph rather than subscribe: AP inbox pull, AP auto-entry, AP statement watch, tune-jobs receipt scan, drop-ship confirmations, letter watch.

5.6 Email (Resend) & SMS (ClickSend)

- **Resend** (`lib/email.ts`): DB-managed keys `RESEND_API_KEY`, `RESEND_FROM`, `RESEND_CAMPAIGN_FROM`, `RESEND_REPLY_TO`, `RESEND_WEBHOOK_SECRET`. If unset, **silently falls back to Graph sendMail**. Delivery/open/bounce webhook: `/api/crm/resend-webhook?key=...`. Per-staff Reply-To comes from `user_profiles.reply_to_email`.
- **ClickSend** (`lib/clicksend.ts`): DB-managed `CLICKSEND_USERNAME`, `CLICKSEND_API_KEY`, `CLICKSEND_FROM`. Returns `clicksend_not_configured` harmlessly until set. Test buttons for both live in the Integrations tab.

5.7 Slack

- **Incoming webhooks** (one env URL per channel): `SLACK_WEBHOOK_URL` plus `SLACK_WEBHOOK_{JAWS_PAYMENTS,VPS_PAYMENTS,JAWS_PAYMENTS,AP_VPS}` and per-tenant B2B order webhook in `b2b_settings`.
- **Bot** (`lib/slack-bot/*`): env `SLACK_BOT_TOKEN`, `SLACK_SIGNING_SECRET`, `SLACK_ALLOWED_CHANNEL_IDS`, `SLACK_PARTS_ONLY_CHANNEL_IDS` (parts-only channels get exactly one stock-availability answer per top-level message, no thread replies, no clarifying questions), `SLACK_PARTS_CONTACT`. Single endpoint `POST /api/slack/ask` handles events, mentions, `/ask`, and interactive buttons; HMAC verified; acks in <3s then continues via `waitUntil`.
- The parts bot answers from `md_stock_cache`, refreshed every 30 min by the `md-stock-sync` GitHub Action.

What we post where. Slack is the business's alerting surface — most automation reports into it rather than email, so a dead webhook is a silent failure. The channels built so far:

Channel	What lands there	Source
<code>#jaws-orders</code>	B2B distributor orders, drop-ship PO confirmations and ETAs	<code>lib/b2b-order-pipeline.ts</code> , drop-ship confirm cron
<code>#jaws-payments</code> / <code>#vps-payments</code>	Payments received, per entity	<code>SLACK_WEBHOOK_{JAWS,VPS}_PAYMENTS</code>

Channel	What lands there	Source
#jaws-invoices / #vps-invoices	Invoice events per entity	webhook
#sales-coaching	Per-call coaching cards, plus the weekly team coaching summary (Mon 07:00)	PBX slack-poster.js
#customer-feedback-negative	Calls flagged as a poor customer experience, with the fault theme. Matt's standing ask is the top-10 recurring faults off this channel. The automation that posts here is currently switched OFF (CALL_CONCERNS_ENABLED).	calls analysis
#customer-feedback-positive	The same in reverse — calls that went well	calls analysis
#parts-room	Parts Office camera motion, nights and weekends only	ja-partsroom on the PBX box
Freight bay	Intrusion bursts from the freight-bay NVR (snapshot sequence)	ja-freightbay on the PBX box — Slack-only since 2026-07-15
AP channel (SLACK_WEBHOOK_AP_VPS)	Supplier-invoice exceptions and suspected double-ups (🔄)	AP auto-entry
#ja-portal-queries	Ask-the-portal bot conversations	lib/slack-bot/*

⚠ Webhook URLs are per-channel env vars. If a channel goes quiet, check the env var before assuming the feature broke — a rotated or deleted webhook fails silently.

5.8 Monday.com & ActiveCampaign (legacy CRM pair — being replaced by the CRM module)

- **Monday** (lib/monday-followup.ts is the shared GraphQL client): env MONDAY_API_TOKEN , MONDAY_BUTTON_SECRET . Five rep quote boards (Dom 5025942308, Kaleb 5025942316, Graham 5026840169, James 5025942292, Tyronne 5025942288). Pipeline A is match-only (never creates items). Button callback: /api/monday/fetch-call-notes?key=... .
- **Quote-channel automations (rebuilt 2026-08-19/20)**: a 3/7/14-day follow-up cadence driven by an FU Stage column (1→2→3; "Follow Up Done" advances the stage, resets Contact Attempts, hides the item in Quote - Pending and re-surfaces it when the date arrives). RLMNA counts attempts: **5 tries in Quote - Lead RLMNA → Quote Not Issued, 3 tries in Quote - Follow up RLMNA → Quote Lost**. After the third follow-up the owner is notified that the decision must be Won or Lost.
 - **Two landmines, both fixed 2026-08-20, worth knowing if the boards misbehave again.** (1) Graham's legacy RLMNA - Step 1 had no group condition and his Follow up RLMNA - Step 1 incremented the counter twice, so one RLMNA click added +3 attempts and lost the quote on the first click; rebuilt as new-builder workflows 1940255686 / 1940255690 with the move ordered *before* the increment. (2) The Follow-

Up-Done handlers are gated on `FU Stage` is exactly 1/2/3, so a **blank** FU Stage makes the click do nothing at all, silently — 132 items (mostly the On Hold cohort the migration skipped) were backfilled to stage 1.

- **MCP cannot delete or deactivate Monday automations** (`USER_UNAUTHORIZED`, even for the account owner — the connector lacks the scope). The workaround is always: create a corrected version, add the bad one to a manual delete list. `create_automation` is also **nondeterministic** — always re-dump with `list_automations` and check the block config; phrase status triggers as "changes to X (from any previous status)" or it will set the from-status to the same label and the workflow can never fire.
- **End of cadence (2026-08-20)**. After the third follow-up the item deliberately does NOT move — it stays in Quote - Follow Up awaiting a Won/Lost decision. Two changes made that legible: a **"Decision Required"** status label the stage-3 handler now sets (workflows 1940257758 Ka / 1940257760 Ty / 1940257765 Ja / 1940257773 Do / 1940257784 Gr), so the row visibly differs from a transient "Follow Up Done"; and the **Owner** column is now stamped by Pipeline A (`REP_BOARDS[] .mondayUserId`). ⚠ **Owner was blank on every item on every board**, so the existing "notify the Owner" automation had been reporting success while notifying nobody — ten of Kaleb's quotes sat at the decision point from mid-June. Historical items were backfilled 2026-08-20 out of hours: **704 active quotes** across the five boards (Tyronne 84, James 188, Kaleb 294, Dom 169, Graham ~111) — scoped to Quote - Pending, Follow Up, Follow up RLMNA and On Hold only. Won, Lost, Not issued and the pre-quote Lead groups were deliberately left blank. Verified: zero blank Owners remain in those 20 groups.
- ⚠ **Never bulk-write to these boards while a rep is working in one**. The item state can change between your read and your write — this bit on 2026-08-20: ten items were read at stage 3, Kaleb reset them to stage 1 and re-ran them seconds later, and the write landed on the new state and had to be undone.
- **ActiveCampaign** (`lib/activecampaign.ts`): env `ACTIVECAMPAIGN_API_URL`, `ACTIVECAMPAIGN_API_KEY`, owner map etc. **Landmine documented in code: `filters[phone]` is a partial match and matches everything on an empty query** — every match must be re-verified by exact digit comparison (this mis-attributed a contact to 26 customers in production once).

5.9 MachShip (B2B freight)

- **Code**: `lib/b2b-machship.ts` + `lib/b2b-freight-carriers.ts` (provider registry — Shippit/Starshipit/AusPost/Sendle slots exist but only MachShip is implemented).
- **Credential**: NOT env — `b2b_freight_carrier_connections` row `provider='machship'`, JSONB `api_token`, header `token`: (not Bearer). Change it in Admin B2B → Settings; effective next call.
- Base `https://live.machship.com` (no sandbox host — test vs live is a property of the token's MachShip user). **Manifesting needs `companyId`, obtained via a consignment GET**. Booking runs with `maxDuration: 120`; consignments that die at MachShip park as `consignment_missing`; `/api/cron/b2b-freight-poll` refreshes status/ETA every 30 min.

5.10 Phones: FreePBX + Deepgram + live monitoring

The portal never calls Asterisk or Deepgram directly — an on-PBX host (CentOS 7, Tailscale `100.82.97.46`) runs four workers and **Supabase is the bus**:

PBX worker	What it does
<code>ja-cdr-sync</code> (<code>sync.js</code>)	CDR → Supabase <code>calls</code> every 5 min (service-role key). Includes park-pickup CDR fix (2026-08-07) and the 6-hour late-arrival lookback (2026-08-21, below).

PBX worker	What it does
<code>ja-transcribe</code> (<code>transcribe.js</code>)	New calls → Deepgram (<code>nova-2-phonecall</code> , <code>en-AU</code>) → <code>call_transcripts</code> . Deepgram key lives on the PBX host.
<code>ja-ami-monitor</code>	Live channel snapshots → <code>POST /api/calls/live/agent/snapshot</code> (~2s, <code>X-Service-Token</code> scope <code>calls:monitor</code>); drains <code>call_monitor_events</code> for Listen/Whisper/Barge and click-to-dial originate.
<code>ja-freightbay</code> / <code>ja-partsroom</code>	Hikvision NVR intrusion events → Slack snapshot bursts + Yealink ring. Node 16 only (glibc). Doesn't touch the portal API.

Call coaching and notes — what this actually gives the business. This is the biggest piece of bespoke work on the phone system, so it is worth stating plainly what it produces:

1. **Every call is recorded and transcribed** — `ja-cdr-sync` lands the CDR, `ja-transcribe` sends the audio to Deepgram (`nova-2-phonecall` , `en-AU`) and stores the transcript. Nothing is manual.
2. **Every call is scored against a rubric for its type** — a sales enquiry, a service booking, a parts call and a pass-off are judged on different things, so the rubrics are per call type (`call_type_rubrics` , editable at Settings → Call Coaching).
3. **Coaching is attributed to the advisor who actually handled it**, identified from the transcript rather than the extension — that is what makes the leaderboard trustworthy when calls get transferred or picked up from park.
4. **Coaching cards post to `#sales-coaching`** per call, and a **team summary posts Monday 07:00**.
5. **Call notes flow back to the quote boards** — the "Fetch Call Notes" button on a Monday item calls `/api/monday/fetch-call-notes` , which pulls that customer's call history and notes onto the item, so a rep picking up a follow-up can see what was last said without hunting for the recording.
6. **Sentiment / objections / conversion** are surfaced on `/calls` as tabs, and calls scoring below 40 are flagged for attention.

The CDR lookback — why it is 6 hours and must stay generous. Asterisk stamps a CDR row with the call's **start** time but does not write the row until the channel hangs up. A long call therefore arrives *behind* the sync watermark. `sync.js` reads `WHERE calldate > (watermark - CDR_LOOKBACK_MINUTES)` ; while that lookback was 30 minutes, **every call longer than about 30 minutes was silently lost or truncated** — the row appeared in MySQL after the watermark had already moved past its start time, so no run ever saw it. Symptom: the portal held no call longer than 38:37 in seven weeks, and a 61:20 call answered by Dom on 204 (2026-08-20 16:16, caller 0428673886) was absent entirely. Fixed 2026-08-21 by raising the lookback to **360 minutes** (`CDR_LOOKBACK_MINUTES` , env-overridable). Re-reading is cheap (~26 rows a run) and safe — the upsert is `on_conflict=linkedid` and the payload carries only CDR-derived columns, so transcripts, recording URLs and coaching analysis on existing rows are never overwritten. **Do not lower it below the longest call the phones can carry.**

Backfilling history. `BACKFILL_FROM` + `BACKFILL_TO` (PBX local time) make a hand-run read exactly that window, leave `sync_state` untouched and skip the recording-upload phase, so it never disturbs the 5-minute timer:

```
BACKFILL_FROM='2026-08-20T16:11:00' BACKFILL_TO='2026-08-20T16:21:00' \
/usr/local/bin/node20 /opt/ja-cdr-sync/sync.js
```

Keep the window tight. A wide one re-upserts every call inside it, and `trg_rescue_missed_call` fires on `UPDATE OF disposition` — which deletes *unread* "Missed call" notifications for the same caller within ± 4 hours of each answered call. There are routinely a couple of thousand unread ones, so a mass re-upsert would quietly clear a slice of them. (`trg_notify_missed_call` is safe: it only fires for calls under 2 hours old.) Note the box's default `node` is v8 and cannot parse the file — the service runs `/usr/local/bin/node20` .

Recordings are transcoded to MP3 before upload. Asterisk records 8 kHz mono WAV at roughly 1 MB per minute, so length translated straight into megabytes and long calls hit the storage ceiling — a 61-minute call was 58 MB and was rejected outright, leaving it with no audio, no transcript and no coaching card. Since 2026-08-21 `sync.js` runs the file through `ffmpeg` (`-ac 1 -ar 16000 -c:a libmp3lame -b:a 32k`) and uploads the MP3 instead: **~75% smaller** (58 MB $\rightarrow$ 14.7 MB), which puts even a multi-hour call comfortably inside the bucket's 100 MB limit. Knobs: `RECORDING_TRANSCODE=0` disables it, `RECORDING_MP3_BITRATE` (default `32k`), `FFMPEG_BIN` , `TRANSCODE_TMP_DIR` , `TRANSCODE_TIMEOUT_MS` .

Details that matter if you touch this:

- **The WAV on disk is never modified** — Asterisk owns it and it stays the source of truth. Only the uploaded copy is compressed, and a failed or empty transcode falls back to uploading the original WAV, so this can never cost a recording.
- **recording_file stays .wav while recording_url becomes .mp3** . They deliberately differ. Anything deciding a content type or a download name must read `recording_url` — that is the object that exists. `transcribe.js` derives the Deepgram content type from it, and both `lib/calls-weekly-report.ts` and `/api/calls/[id]/recording-url` were corrected to do the same.
- Old `.wav` objects stay exactly as they are and still play; the portal's `<audio>` element and Deepgram both handle either format.
- `transcribe.js` falls back to inserting the transcript **without raw_response** if the full insert fails. That column holds the entire Deepgram document with word-level timings (~940 kB on a 61-minute call) and nothing reads it, so a statement timeout on the payload must not cost the transcript — which is exactly what happened on the first 61-minute call before the fallback existed.

Analysis runs on the PBX (`/opt/ja-cdr-sync/analyse.js` + `slack-poster.js`); the portal's `/api/cron/calls-analyse` equivalent is **gated off by default** (`CALLS_ANALYSIS_ENABLED`) and must stay off while the PBX loop runs — both running means double-scoring and double-posting. Browser softphone (SIP.js over WSS) config is env `NEXT_PUBLIC_FREEPBX_WSS_URL` etc.; TURN needed on mobile networks.

5.11 Anthropic / Claude API

Env `ANTHROPIC_API_KEY` ; raw fetch to `/v1/messages` . 18 call sites: AP extraction & statement parsing, quote PDF extraction, call analysis/coaching/concerns/weekly report, follow-up summaries, workshop-map weekly narrative, B2B drop-ship confirmation reading, training-course generation, tune-job receipt extraction, report narratives, sales-recap flags, Slack bot, portal chat. Most models overridable per-feature via env (`*_MODEL` vars).

5.12 Everything else

- **Web Push**: `NEXT_PUBLIC_VAPID_PUBLIC_KEY` / `VAPID_PRIVATE_KEY` (watch for whitespace in env values — known gotcha); dormant no-op when unset.
- **MCP connector**: portal is a read-only MCP server at `/api/mcp` (personal `jap_` tokens from Settings, or OAuth 2.1 with dynamic client registration).

- **Google Places** (`NEXT_PUBLIC_GOOGLE_PLACES_API_KEY` , client-exposed — must be referrer-restricted) for workshop address autocomplete; **Nominatim** (keyless, throttled ~1 rps) geocodes tune-job addresses; an offline AU postcode dataset powers the workshop map.
- **GitHub dispatch**: `GH_DISPATCH_TOKEN` (`actions:write`) lets portal routes fire `repository_dispatch` events to run MD workers on demand.
- **Website lead intake**: `POST /api/crm/intake` guarded by `x-crm-token` = DB-managed `CRM_INTAKE_TOKEN` (rotate from the Integrations tab).
- **CData MCP**: decommissioned 2026-07-14. `lib/cdata.ts` is dead code; health check hardcoded green/deprecated.
- **Base URL sprawl**: seven overlapping env vars (`NEXT_PUBLIC_APP_URL` , `NEXT_PUBLIC_SITE_URL` , `NEXT_PUBLIC_BASE_URL` , `NEXT_PUBLIC_PORTAL_URL` , `PORTAL_PUBLIC_URL` , `APP_BASE_URL` , `JA_PORTAL_BASE_URL` , `B2B_PUBLIC_BASE_URL`) — keep them consistent; mismatches silently break OAuth redirects and email links.

6. Scheduled automation

6.1 Vercel crons (25 — `vercel.json` ; schedules are UTC; Brisbane = UTC+10)

Cron	Brisbane time	Purpose
<code>/api/distributors/refresh-cache</code>	02:00 daily	Warm <code>distributors_cache</code> (FY figures + last 13 months)
<code>/api/cron/sync-followups</code>	every 5 min	Call follow-ups → Claude summary → AC contact + Monday push (<code>FOLLOWUP_SYNC_ENABLED</code> kill switch)
<code>/api/cron/calls-analyse</code>	every 5 min	Portal-side call coaching sweep — off by default (<code>CALLS_ANALYSIS_ENABLED</code>); PBX box currently does this
<code>/api/cron/calls-weekly-report</code>	Mon 07:00	Weekly coaching narrative → <code>#sales-coaching</code>
<code>/api/cron/workshop-map-weekly</code>	Mon 07:10	Quotes/jobs geography report → email Matt (cc Ryan, Chris). Its <code>md_quotes/md_invoices</code> reads are <code>PAGED</code> (<code>selectAllRows</code>) — a plain select hits the 1000-row cap and under-reports
<code>/api/cron/renew-graph-subscriptions</code>	every 6 h	Extend Graph mailbox subscriptions expiring <24h
<code>/api/cron/health-check</code>	every 5 min	The 21 integration health checks → <code>integration_health</code>
<code>/api/cron/b2b-freight-poll</code>	every 30 min	MachShip status/ETA poll on open consignments
<code>/api/cron/workshop-reminders</code>	every 15 min	Queue + send service/regio reminder SMS (respects <code>sms_enabled</code>)
<code>/api/cron/notifications-sweep</code>	every 15 min	Bell notifications for Monday-side events (new leads, new to-dos)

Cron	Brisbane time	Purpose
/api/cron/b2b-payment-check	every 6 h	Confirm BECS settlement in MYOB → mark <code>payment_settled_at</code>
/api/cron/crm-automations	every 5 min	CRM automation flow engine
/api/cron/task-automations	every 5 min	Tasks automation flow engine
/api/cron/crm-campaigns	every 5 min	Campaign scheduler + Resend queue drain + call linkage
/api/cron/ap-statement-watch	every 10 min	Supplier statements → reconcile vs MYOB → digest email (report-only)
/api/cron/ap-auto-entry	every 15 min	VPS inbox → fact-check → auto-post clean invoices to MYOB (<code>AP_AUTO_ENTRY_ENABLED</code>)
/api/cron/overnight-leads-snapshot (+morning variant)	every 30 min / 5 min 06–08	Snapshot Monday lead groups for the sales recap
/api/cron/letter-watch	hourly	New finalised MYOB VPS invoices → thank-you letter + envelope print jobs (deposit-only invoices vetoed)
/api/cron/agents	every 15 min	Monitoring agents framework (<code>lib/agent-framework</code>)
/api/cron/tune-jobs	hourly :20	Scan inbox for Stripe tune receipts; Monday-morning reminders/escalations
/api/cron/b2b-dropship-confirm	every 15 min	Supplier confirmation emails → full drop-ship receiving flow
/api/cron/mgmt-dashboard-warm	05:30 daily	Pre-compute Management Dashboard MYOB bundles
/api/cron/leave-decisions	every 15 min	Leave applications decided on the Monday board → email the applicant (approved <i>and</i> denied), cc/reply-to HR. Only items on the application path count — see §7.17. <code>LEAVE_EMAILS_ENABLED=false</code> switches it off
/api/cron/jaws-stock-eom	07:30 on the 2nd	JAWS month-end stock report for the month just ended → snapshot (migration 199) + email to <code>JAWS_EOM_EMAIL_TO</code> . Cron can't express "last day of month"; running into the new month also guarantees the month's invoices are all in. See §7.18

⚠ Two handlers exist but are **not scheduled** (headers claim otherwise): `bank-payments-slack.ts` (7am payment digests) and `slack-cleanup.ts` (parts-bot TTL deletes). Manual-invoke only until added to `vercel.json` .

6.2 GitHub Actions (Mechanics Desk workers — 16 workflows)

MD has no API; these Playwright workers log in with `MECHANICDESK_{WORKSHOP_ID,USERNAME,PASSWORD}` secrets and talk back to the portal with `X-Service-Token: $JA_PORTAL_API_KEY` . All install Chromium at run time (`setup-node@v5` + `npm playwright install chromium` — the old "run in the MCR container" note in

05_INTEGRATIONS.md is obsolete). **playwright** in devDependencies must match the installed version (1.59.1). MD allows a single session per employee — concurrency groups prevent workers evicting each other.

Workflow	Schedule (Brisbane) / trigger	Job
mechanicdesk-pull	08/10/12/14/16:00	MD WIP report → forecast ingest
md-stock-sync	every 30 min, 06:00–19:30 Mon–Sat	Full MD stock catalogue → md_stock_cache (Slack parts bot)
md-workshop-map	03:30 nightly	Full invoices/quotes/customers pull → workshop map (FY2025 was a one-time backfill; nightly FROM = 2025-07-01)
mechanicdesk-prepick	06/11/15:00 weekdays	Diary-job parts demand → Pre Pick snapshot
sales-recap	Mon 07:00 full+email; ~7 intraday MD refreshes	Weekly Sales Recap (Monday boards + MD scrape → email Ryan)
md-customer-import	02:00 nightly	Monday quote-channel leads → MD customers (MD can't search phones)
tune-jobs-md	02:30 nightly	Submitted tune jobs → MD customer + vehicle + note
mechanicdesk-stocktake	repository_dispatch	Stocktake match / push / recheck / refresh / push-missing
md-purchase-order	repository_dispatch	Raise MD PO after a JAWS→VPS stock transfer
7 × probe-md-* / fix-md-customer-names	manual	Endpoint recon probes and one-off repairs

6.3 The print centre (comms room)

PORTAL-CENTRE is a dedicated always-on Lenovo ThinkCentre in the comms room whose only job is printing. It exists because the print agent used to run on Chris's MSI laptop, which meant **every letter, invoice and label silently stopped whenever the laptop was closed or off the network** — with no error anywhere, because the queue simply stopped draining. Built and verified 2026-08-20.

Host	Lenovo ThinkCentre, comms room, always on (no lid, no sleep)
Reachable	Tailscale 100.72.189.95 — RDP in for maintenance; it is Entra-joined, so sign in with the work account
Runs	agents/label-print-agent/ on Node 22, started by Autologon + Startup shortcut, headless
Queue	label_print_jobs in Supabase, consumed over Realtime with a 30s poll as fallback
Job kinds	label (DYMO 4XL), invoice, letter, envelope (Apeos) — all four verified end to end
Runbook	docs/print-centre-thinkcentre-setup.md — build, headless operation, the comms-room move sequence, RDP, and a health-check script

Operating notes

- The agent **never restarts on deploy** — it is not part of the Vercel app. A new print kind must not require an agent change; add it as a queue row shape the agent already understands.

- A stuck queue drains on agent restart, so restarting the agent is the first fix. Failed jobs are re-queued by flipping `failed` → `pending`.
- ⚠ **Never trust the Apeos (Copy 1) printer-name suffix.** Windows appends it when a printer is re-added, and printing to the wrong name fails silently or prints to the wrong device — check the real name.
- Printer enumeration must never take the heartbeat down (fixed 2026-08-20) — if the agent looks dead but the box is up, check the heartbeat before rebuilding anything.
- DYMO troubles: the health-check script has label-printer diagnostics and a `-FixDymoPort` switch.

6.4 On-premise agents

Agent	Where	Notes
<code>label-print-agent</code> (<code>agents/lab-el-print-agent/</code>)	PORTAL-CENTRE — see §6.3	Consumes <code>label_print_jobs</code> via Supabase Realtime (30s poll fallback). Kinds: <code>label</code> (DYMO 4XL), <code>invoice</code> , <code>letter</code> , <code>envelope</code> (Epson). Never restarts on deploy — new print kinds must not require agent changes. Runs via Startup shortcut → <code>run-agent-hidden.vbs</code> or NSSM. Node 22, Autologon, reachable on Tailscale <code>100.72.189.95</code> ; built 2026-08-20, runbook <code>docs/print-centre-thinkcentre-setup.md</code> . Previously ran on Chris's MSI laptop and failed silently whenever it was off-network — that is what the dedicated box fixes. Never trust the Apeos (Copy 1) printer-name suffix.
PBX workers	FreePBX box	See §5.10. systemd units; Node 16 for the camera bridges.
ESP32 scale nodes	Workshop bins / cash-count rig	Firmware <code>hardware/ja-scale-node/</code> ; raw HX711 counts to <code>/api/scales/ingest</code> ; all calibration server-side.

7. Modules & SOPs

Full route-by-route inventory: 115 page routes (435 API routes). Staff nav is the `/home` app launcher (role- and `visible_tabs` -filtered). Below, per module: what it is + how to operate it.

7.1 Dashboards (`/dashboard` , `/overview` , `/home`)

Section-based management dashboard (JAWS+VPS merged, entity filter pills) and a drag/resize custom dashboard builder with per-user saved layouts. **SOP:** nothing operational — data flows from MYOB reporting caches; if figures look stale, check the 02:00 `distributors/refresh-cache` cron and the health page.

7.2 Workshop tooling (Mechanics Desk stays the system of record)

Mechanics Desk is the workshop system. The diary, job cards, customers, vehicles, quotes and invoicing all happen in MD, and there is **no migration in progress** (decision 2026-08-20). The portal does not run the workshop.

What the portal *does* provide around MD, all of it in daily use:

Surface	What it does	Fed by
Letters (<code>/workshop/letters</code>)	Hourly <code>letter-watch</code> cron queues a thank-you letter + envelope for each newly finalised MYOB invoice and prints them on the Apeos. Deposit-only invoices are vetoed by a positive 1-1230 line. Reprint / compose / edit templates here.	MYOB (not MD)

Surface	What it does	Fed by
Pre Pick (/workshop/prepick)	14-day parts demand vs stock on hand.	mechanicdesk-prepick Playwright worker
Purchase Orders (/workshop/purchase-orders)	PO draft → sent → received, with a bill push to MYOB.	Portal + MYOB
Stocktake (MD) (/stocktake)	Count sheet reconciled against MechanicDesk.	MD workers
Stocktake (Portal) (/workshop/stocktake)	Barcode session → count → Variance → Apply.	Portal
Stock Transfer (/admin/b2b/stock-transfer)	JAWS↔VPS paired invoice + bill, plus the MD PO.	Portal + MYOB + MD

The parked sections are switched off in the UI (2026-08-20). `lib/workshop-sections.js` is the single source of truth for which sections are on; the tab strip reads it, and `next.config.js` reads the same list to rewrite the parked routes to a "not in use" notice.

- Off: Diary, Jobs, Customers, Vehicles, Quotes, Invoices, Comms, Workshop Reports (plus their `/workshop/{job,customer,vehicle,quote,invoice}/{id}` detail routes), and — second wave, same day — Orders, Inventory, Live Bins, Cash Count, Suppliers. 18 routes in total.
- On: the six in the table above.
- **The nav is now a single flat strip.** `components/InventoryTabs.tsx` (the old Inventory second level) was deleted once Inventory/Live Bins/Cash Count/Suppliers/Orders came off — there was almost nothing left to wrap — and its survivors were promoted to the top strip.
- The Workshop app tile opens `/workshop/prepick`.
- Parked routes **rewrite**, not redirect — the URL stays, so an old `/diary` bookmark still reads `/diary` and explains itself. The rewrites are in `beforeFiles`; an array return or `afterFiles` is evaluated *after* the filesystem and the real page would render instead.
- `?preview=1` on a parked route skips the rewrite and loads the real page — an escape hatch for anyone reviving the build.
- **To switch a section back on: flip `active` to `true` in `lib/workshop-sections.js`.** Nothing else. `docs/workshop_md_parity.md` still holds the parity and cutover detail.

⚠ Because MD stays, the **9 scheduled MechanicDesk scrapers remain a live production dependency** (\$6.2), not a temporary bridge. Treat them accordingly — an MD password change or UI change breaks real reporting.

7.3 B2B — distributor portal (/b2b/*)

Distributor experience: Shop → Cart (PO number required, ≤20 chars — MYOB limit) → Stripe Checkout (card+surcharge / PayTo / BECS) → Orders (status timeline + freight tracking) → Jobs (tune receipts to fill in) → Resources → Training → Team → Settings. Design language: the Alloy kit (`components/b2b/ui.tsx`) — distributor portal refreshed 2026-08-12, staff admin brought onto the same kit 2026-08-20.

Pricing is GST-INCLUSIVE on every distributor-facing surface — catalogue tiles, cart lines, cart totals, freight options, order list, order detail and the order emails. Prices are *stored* ex-GST (`unit_price_ex_gst` , `trade_price_ex_gst` , `price_ex_gst`) and converted for display by a local `incGst(ex, taxable)` helper in `catalogue.tsx` and `cart.tsx` — taxable items +10%, FRE items as-is, which is why the flag matters and a blanket $\times 1.1$ would be wrong. Audited end to end 2026-08-25; the one surface that still led with an ex-GST figure (the order-detail totals block, "Subtotal (ex GST)") now reads *Items (inc GST) / Freight (inc GST) / Total paid* with an "Includes \$X GST" line, matching the cart. **Freight is folded into `subtotal_ex_gst` at checkout**, so the order detail recovers it as the remainder — `items inc + freight inc == subtotal_ex_gst + gst` — rather than re-taxing `freight_cost_ex_gst` at a fixed 10%, which would be wrong for any non-taxable line. Staff-facing admin surfaces (freight zones, dropship calibration, catalogue audit/export) deliberately stay ex-GST: that is how the costing is done, and they are labelled. The **tax invoice PDF** also stays ex + GST + total, as an ATO tax invoice must.

What happens automatically on payment (`lib/b2b-order-pipeline.ts`): order → paid; cart cleared; MYOB invoice written cent-exact inc-GST; consignment-first pick list printed; for drop-ship items a supplier PO is created and emailed; Slack notification; freight bookable via MachShip (admin button or login-less email link).

The tax invoice is NOT raised at booking — see Freight & despatch below.

SOPs

- **Onboard a distributor:** Admin → B2B → Distributors → Add (live MYOB customer typeahead links the card) → open the distributor → Invite users (magic-link / welcome token email). Owners manage their own team after that.
- **Over-limit orders:** quantities above `over_limit_qty` route to a quote-or-dropship flow instead of straight checkout.
- **Bundles:** "includes" children ship inside the parent's box (affects freight cartonization).
- **Refunds:** admin order page → Refund modal → **Items mode** (pick lines + quantities; amount derived server-side from checkout-exact pricing; `refunded_qty` prevents double refunds) or amount modes (covers freight/surcharge). Mirrored to MYOB as a credit note with negative item lines (Xero path posts on B2B_SALES).
- **Drop-ship receiving:** supplier replies to the PO email → `b2b-dropship-confirm` cron reads it → auto-bills the PO, flips the sale order to invoice, receipts payment, relays ETA to the distributor, posts to `#jaws-orders`. Manual MYOB conversions are adopted rather than duplicated.
- **Freight & despatch (changed 2026-08-20 — read this):** booking and despatch are now two separate steps.
 - **Book Freight** only *prepares*: it creates the MachShip consignment (left **Unmanifested**) and prints the pick slip + consignment note/labels so the order can be picked and packed. Nothing reaches the carrier at this point and no tax invoice is raised.
 - **Ship Now** (`lib/b2b-ship-now.ts`) is what actually despatches: manifests the consignment, converts the MYOB Sale.Order → Sale.Invoice, receipts the payment against it, prints the A4 tax invoice, emails/pushes the distributor and stamps `shipped_at`.
 - Bulk despatch is deliberately **one manifest, not N** — MachShip's manifest call also books a carrier pickup window, so manifesting ten consignments individually would raise ten pickup requests. Select the run and ship it in one action.
 - This reverses the 2026-08-06 behaviour where booking manifested immediately. If nothing is reaching the carrier, the likely answer is simply that nobody has pressed Ship Now.

- Rates: per-zone + flat-rate satchels (weight-gated; satchel rows may need seeding) + drop-ship per-zone rates; calibration panel at Admin → B2B → Dropship Calibration. If a consignment goes missing at MachShip it parks as `consignment_missing` — rebook from the order page.
- **Tune jobs:** Stripe receipt lands in the accounts inbox → hourly cron extracts it → distributor fills customer/vehicle at `/b2b/jobs` (or the login-less weekly reminder link) → nightly GH Action creates the MD customer/vehicle/note; Monday item + thank-you letter follow. One job per VIN. Staff-side management (aliases, retries, dismiss) at Admin → B2B → Tune Jobs.
- **Training:** Admin → B2B → Training → assign per distributor or per user (assignment-gated). Courses can be generated from an uploaded PDF (LLM pipeline). Edit quiz answers at `.../training/[slug]/answers`; preview renders the real player without recording attempts.
- **Testing safely:** Admin → B2B → Test Order exercises the full real pipeline against a chosen distributor.

7.4 B2B — staff admin (`/admin/b2b/*`)

Dashboard · Catalogue (inline price/visibility edits + drawer) · Stock Order (reorder forecasting, replaces the JAWS Excel) · JAWS Stocktake (count-sheet vs MYOB on-hand, **report-only**) · Stock Wall (saved on-hand tile views; also what suppliers see) · Stock Transfer (JAWS↔VPS paired invoice+bill + MD PO) · Distributors · Suppliers (logins) · Orders · Tune Jobs · Resources (sectioned doc library, signed uploads) · Training · Settings (Stripe status, freight carriers/zones/packaging, sender address, email templates).

7.5 Accounts Payable (`/ap` , `/ap/[id]` , `/ap/statement`)

Supplier emails → Graph inbox pull → Claude extraction → triage list.

Daily SOP (Amanda): `/ap` → Pull from Inbox if needed → review each invoice (`/ap/[id]` : PDF preview, line editor with account suggestions, MD job link, supplier presets) → Approve (pushes header + lines to the right MYOB entity) or Reject. Green-triage rows support bulk approve.

Automation: `ap-auto-entry` cron (VPS, gated by `AP_AUTO_ENTRY_ENABLED`) posts clean invoices automatically and Slacks a breakdown; supplier allowlists control consolidated and pay-on-proforma handling; duplicates get a 🔄 Slack and are filed to Read/Printed; **locked-period invoices are flagged, never auto-redated**; supplier matching is suffix-blind; link-only emails (no PDF attached) are invisible to the pipeline. `ap-statement-watch` cron reconciles statement PDFs against MYOB and emails a digest (report-only; Capricorn statements are report-only by policy). Manual statement recon UI: `/ap/statement` .

The Slack flag card is the human interface to the automation (`lib/ap-auto-entry-slack.ts` , clicks handled in `/api/slack/ask`). Each flagged invoice carries a row id in `ap_auto_entry_log` and up to four buttons: *View invoice* (7-day signed URL),  *Approve & post to MYOB* (`approveAndPost` — re-extracts with the strong model, soft checks bypassed),  *Create supplier* (`proposeSupplier` → threaded review → `approveCreateSupplierAndPost`), and  *Entered manually?* (`checkEnteredManually`). JAWS cards additionally get an account-choice row (`postWithAccount`).

"Entered manually?" (migration 200 , 2026-08-25) closes the automation's blind spot: staff key flagged invoices into MYOB by hand and the card stays orange forever. The check is read-only against MYOB, in three widening nets: (1) same supplier + same `SupplierInvoiceNumber` via `findExistingMyobBill` (already OCR-tolerant and amount-aware); (2) that invoice number under **any** supplier — the flag often exists *because* no card matched — adopted only when the amount agrees, so a number collision across suppliers can't link the wrong bill; (3) recent bills for the supplier at the same amount under a different number, which are **not** adopted silently but returned as threaded candidates with a  *Link bill #...* button (`linkManualBill`). A hit sets `outcome='posted'` , `entered_manually=true` , `manual_checked_by/at` , links `myob_bill_uid` , files the email

away, and flips the card to  *Posted manually* (`markPostedManuallyBlocks`). `entered_manually` rows are excluded from the supplier-trust counts — they're evidence a *person* handled the supplier, not the automation. Nothing is ever posted to MYOB by this path. Cards posted before the button existed can be retro-fitted in place (chat.update, so the card keeps its ts and thread) with `GET /api/ap/admin/backfill-manual-button?days=N&dry=1` (`edit:supplier_invoices` , `backfillCheckManualButtons`); it only touches still-open flags and skips cards that already carry the button, so it is safe to re-run.

Gotcha, fixed 2026-08-25. Slack stores the emoji you post as its **shortcode**, so a card read back through `conversations.replies` (or arriving on a button's interaction payload) has the header `:large_orange_circle:` Not auto-posted – Supplier, *not* the  ... that was sent. The first backfill run matched the literal emoji and therefore skipped all six cards it was pointed at — silently, because the skip reason was a catch-all. `OPEN_FLAG_HEADER` now accepts both forms and anchors on the words "Not auto-posted"; `markApprovedBlocks` shared the same flaw (an approved card's header would read "  Not auto-posted — ...") and is fixed by the same regex. Skip reasons are now specific and quote the header they saw.

Migration 200 also fixed constraint drift found while building it: `lib/ap-auto-entry.ts` has written `outcome='skipped_duplicate'` since the cross-source duplicate guard shipped, but the check constraint from migration 145 never listed the value — so **every one of those audit rows was silently rejected** (0 rows in the table) and the attachment was re-processed, and re-extracted, on the next run. The value is now allowed.

7.6 CRM (/crm/*)

Pipeline kanban, contacts (+timeline), campaigns (Resend), React Flow automations. Replaces Monday quote boards + ActiveCampaign + Zapier — **manual cutover steps are recorded in the project memory/notes; AC + Monday remain live in parallel until cutover.** Website leads arrive via `/api/crm/intake` (token-guarded). Three crons drive automations/campaigns/call-linkage every 5 min.

7.7 Calls (/calls)

CDR list with audio, transcripts, coaching analysis (per-call-type rubrics), Sentiment/Coaching/Words/Conversion tabs, live Listen/Whisper/Barge (needs `monitor:calls`), click-to-dial (`use:phone` , `NEXT_PUBLIC_CLICK_TO_DIAL=1`). Coaching attribution keys on the transcript-identified effective advisor. Weekly coaching report posts to Slack Monday 07:00. **Negative-call (concerns) automation is fully built but switched OFF** — `CALL_CONCERNS_ENABLED=true` resumes it.

SOP if calls stop appearing: check Connections page (`freepbx_cdr_sync` freshness) → SSH to the PBX via Tailscale → check `ja-cdr-sync` systemd timer. Transcripts stale → `ja-transcribe` . Live monitor "not configured" → `ja-ami-monitor` hasn't pushed a snapshot in >20s.

7.8 Reports (/reports/*)

Sub-tabs: Reports · Sales Report · Sales Dashboard · Management Dashboard · Forecast · Workshop Map · Distributor Map · **Stock EOM** (\$7.18, `view:stock`) · Distributors.

Builder (6 PDF report types — Workshop Performance was removed 2026-08-20; it reported over the portal workshop tables, which stay empty while MechanicDesk is the system of record) · Sales Report (live Weekly Sales Recap; "**sales**" = **orders taken from Monday boards + MD, not turnover**; auto-emails Ryan Mon 07:00) · Management Dashboard (JAWS weekly Excel replica from live MYOB; config-driven charts; clickable KPI history; cache warmed 05:30) · Workshop Map (nightly MD pull; `lib/workshop-map` classification is authoritative; FY

picker; five tabs — Jobs Map, Quotes Map, Conversion, By State, **Vehicle Trend**: one line per vehicle series, All FY = monthly buckets, pick a month = daily buckets, measures Jobs/Quotes/Job \$/Quoted \$. The trend counts every invoice and quote, so its totals run higher than the map tabs, which show one dot per customer per month) · Distributor Map (quotes near each distributor vs confirmed Monday bookings). Both maps export the full FY month by month as a PDF — see below · **Sales Dashboard** (daily/monthly/total sales taken, plus a quote-pipeline view — see below) · **Forecast** (admin+manager; portal rebuild of the Monday "Forecast Dashboard - Includes JAWS" — see below). Per-user report-tab allowlists control who sees what.

Just Autos on the Distributor Map (2026-08-25). `lib/distributor-map.ts` appends a synthetic entity — key `ja:home`, name "Just Autos (workshop)", located from postcode `4560` (Burnside; override with `DISTRIBUTOR_MAP_HOME_POSTCODE`) — flagged `quotesOnly: true`. It is pushed **after** the Monday booking pass and **before** the quote pass, deliberately: it must compete for quotes on the same nearest-within-radius rule, but must never bind a Monday label (adding it earlier would put it in `names` and shift the indices `matchLabel` returns).

`quotesOnly` is a contract, not a hint — every consumer must honour it: booked/capture render as `-` not `0`, it is excluded from the "All distributors" totals in both the dashboard and the PDF, and `distributorAreasForMonth` filters it out so it never reaches the weekly sales recap. Fold it into a total and the combined capture rate drops without any distributor having done worse.

Geometry worth knowing: the nearest distributor to Burnside is ~390 km (Banana Coast, Coffs Harbour), and the radius selector offers 50/100/150/250 km — so below 250 km the new entity cannot take a quote from anyone; at 250 km the areas overlap and quotes nearer the workshop move to it. The pre-existing `DISTRIBUTOR_MAP_EXCLUDE` filter (default `just autos`) still guards the `b2b_distributors` query; there is no JA row there today, so it is defensive only and unrelated to this entity.

Map PDF exports (added 2026-08-25). Both map reports have an **Export PDF** button that prints the whole FY month by month — the screen shows one month at a time and there was no way to take a year off it.

	Workshop Map	Distributor Map
Renderer	<code>lib/workshop-map-pdf.tsx</code>	<code>lib/distributor-map-pdf.tsx</code>
Route	<code>GET /api/workshop/map/pdf?fy=&cat=&state=</code>	<code>GET /api/reports/distributor-map-pdf?fy=&radius=</code>
Data	the cached <code>md_workshop_map_cache</code> payload — same one the screen reads, so it returns in ~1s and never triggers an MD pull	recomputes via <code>computeDistributorMap</code> (Monday is live), ~seconds, <code>maxDuration</code> 120
Gate	<code>view:reports</code>	<code>view:reports</code>

Both follow the house `@react-pdf/renderer` style of `lib/jaws-stock-eom-pdf.tsx` (A4, Helvetica, fixed table headers that repeat on page breaks, `wrap={false}` rows). Neither is month-filtered by design — the month strip narrows the screen, not the export; the vehicle/state (workshop) and radius (distributor) filters *do* carry through, into the content and the filename.

Two populations in the workshop PDF, and they are not interchangeable. Counts come from `payload.conv` (every deduped record in the FY); revenue comes from `payload.*.points` (geocoded records only — an ungeocoded job carries no amount in the payload at all). Every table says which it is using and the last page prints the coverage. A "total revenue" that mixed them would be wrong, which is why they are kept apart rather than added.

`pcState()` (postcode → state) moved out of `WorkshopMapDashboard` into `lib/workshop-map/postcode-state.ts` so the dashboard and the PDF classify identically — they previously would have held separate copies.

Weekly Quotes & Jobs Map Report (`lib/workshop-map-weekly-report.ts` , cron `/api/cron/workshop-map-weekly` , Mon 07:10). Aggregates the previous 7 days of `md_quotes` / `md_invoices` by locality and vehicle group against a prior-4-week baseline, has Claude write the “what it means / where to market” read, and emails Matt (cc Ryan, Chris). `?dry=1` returns the JSON without sending; `?days=N` widens the window.

Gotcha, fixed 2026-08-24. Both fact-table reads used a big `.limit()` , which PostgREST ignores — it caps responses at 1000 rows and drops the rest **silently**, off the end of the sort order. As the 5-week quote window grew past 1000 rows the newest week was the part that got truncated, so the report undercounted for weeks (222 reported vs 290 actual on 16 Aug) and then read a flat **“0 quotes issued” against 313 real quotes** on 24 Aug. Both reads now page via `selectAllRows()` ordered on the tables’ primary keys. The failure mode to remember is that it produces a *plausible smaller number*, not an error.

Sales Dashboard (`/reports/sales-dashboard` , added 2026-08-21). The portal rebuild of the Monday **“Sales Dashboard”** (2079976). `view:reports` , same as the Sales Report. **Three** views behind the one tab, each fetching only when first opened — this is also where the Monday **“Management Dashboard”** (321206) was rebuilt, per Chris 2026-08-21:

1. Management (default) — the Monday Management Dashboard, widget for widget: sales orders vs target **per month** (last 18), **per salesperson** (current month) and **per day** (last 30), plus the **Cancelled** and **Postponed** order totals.

Each chart keeps its own period because each target is stated per that period — the giveaway was that the five reps on Monday’s P/P chart sum to roughly ONE month’s total, so it is a monthly per-person target, not a running one.

- **Targets** resolve DB-first through `lib/integration-config`: `SALES_TARGET_PER_MONTH` / `SALES_TARGET_PER_PERSON` / `SALES_TARGET_PER_DAY` , defaulting to the values read off Monday’s target lines (**\$1,000,000 / \$300,000 / \$50,000**). Changeable without a deploy.
- **Cancelled and Postponed are whole-board GROUP totals with no date filter**, and the group is the authority, not the status column — the Postponed group holds rows whose status says “Done” or “Not Done” and they still count. Verified to the cent on 2026-08-21: the 12 rows in “Postponed Orders” sum to **\$73,804.45**, exactly what the Monday widget shows. `?exceptions=1` on the API opts into this extra pull; only this view needs it. “Cancelled Orders - Previous Years” is a separate group, **excluded by default** — `SALES_EXCEPTIONS_INCLUDE_PREV_YEARS=1` folds it in.
- These exception totals are **excluded from every sales figure** — cancelled and postponed work is not revenue. They sit alongside as what was lost and what is parked.
- **“Staff Parts Owing” is NOT built.** There is no board of that name (the dashboard names 9 connected boards), so it must be a filtered view on one of them; the tile renders as “—” with a note rather than inventing a number. Reads \$0 in Monday today.
- Per-salesperson excludes **Unassigned** so the rep comparison isn’t distorted by rows with nobody set; those are reported on the Figures view instead.

2. Figures — what the Monday dashboard actually tracks: **daily, monthly and per-salesperson sales taken over any date range**. `lib/sales-figures-monday.ts` + `GET /api/reports/sales-figures?start=&end=&months=&days=&person=` . An explicit `start / end` wins over `months` ; a backwards range is

swapped rather than returning nothing. Built on `fetchOrders / fetchDistBookings from lib/sales-recap-monday` rather than re-pulling the boards, so the definition of "a sale" lives in ONE place — those already drop the dead statuses (Deleted / Canceled / Cancelled) and the Distributor-Booking groups that don't count (Booking - Pending, Postponed...). Change it there and both this and the Weekly Sales Recap follow. Daily fills every calendar day so quiet days and weekends read as gaps rather than closing up the axis; the average is per **trading day** (days with at least one sale), because dividing by calendar days quietly halves it. Best day and best month are period-wide, not limited to the daily window. Month/year-to-date return 0 when the range ends before today, rather than looking like a fault.

Salesperson attribution. The people column — "Created By" on Orders, "Person" on Distributor - Booking, both id `person` — was added to `ORD / DB` and to `OrderRow / DistRow` in `lib/sales-recap-monday` for this (additive; the Weekly Sales Recap and Distributor Map ignore it). A row can name **several** people; it counts against the **first** only, so the per-person rows still add up to the period total — counting a shared row against everyone named would inflate the total and make the table irreconcilable with the headline. Rows with nobody set group as **Unassigned** rather than being dropped. `?person=` scopes the tiles, charts and sale-type split; the per-salesperson table always covers the whole range so you can still see everyone while filtered.

3. Pipeline — the quote pipeline across the five rep Quote Channel boards. `lib/sales-dashboard-monday.ts` + `GET /api/reports/sales-dashboard?months=3|6|12|24`. Open pipeline by stage, won/lost by month, a per-rep table with win rate, and an ageing breakdown.

Sales taken ≠ turnover. Both views count orders/bookings *placed*, the same meaning the Sales Report uses; invoiced turnover is Reports → Forecast. The two will not agree and are not meant to.

Three things the Pipeline view has to get right:

- **Group ids drift per board — resolve by TITLE, never by id.** Only the five original template groups (`topics` , `group_title` , `new_group__1` , `new_group` , `new_group860`) are shared across the boards. "Lead RLMNA", "Follow up RLMNA", "On Hold" and "Not issued" were added after the fork and every board has its own ids — Kaleb's On Hold is `group_mm12crrx` , Dom's is `group_mm12q0j0` . **The GROUPS map in `lib/monday-followup` holds one board's ids and is correct for the shared five only — do not reuse it for cross-board reporting.**
- **Rep comes from the board, not the Owner column.** The Owner backfill of 2026-08-20 covered the ~704 *active* quotes; historical Won/Lost items have an empty Owner, so board-as-rep is the only attribution that carries history.
- **Open is defined by exclusion** — everything that is not Won, Lost or Not issued. That way a group nobody mentioned still appears instead of vanishing: Kaleb has a "Farm Fest 2026 booked in Jobs" group, which is counted in the pipeline *and* listed separately on the page so the totals can always be reconciled to the boards.

The columns it reads *are* shared across all five boards (verified 2026-08-21): `numeric_mkzcbhz2` Quote Value (titled "Value" on James's board, same id), `date4` , `status` , `person` . The ones that drift — Distributor, Qualifying Stage, Contact Attempts, FU Stage, Quote No — are not read. Monday does the group and date filtering server-side, so the ~6,500 items across the boards cost one group-resolve plus two queries per board.

Forecast (`/reports/forecast` , added 2026-08-21). The portal rebuild of the Monday **"Forecast Dashboard - Includes JAWS"** (dashboard 349826), which sits on the Monday **"Forecasting"** board `1842188200` .

`lib/forecast-monday.ts` pulls the board live on every load — 24 items, one Monday query, nothing cached or stored — and `GET /api/reports/forecast` serves month × entity × year turnover. Admin + manager only, same as the Management Dashboard.

Three things about that board that the parser has to defend against, all confirmed against live data:

- **Item names are unreliable, so the month always comes from the GROUP.** The December group contains an item titled "November Job Report - JAWS", March's is "Mach Job Report" (typo), and August's and September's are simply "JAWS". Reading the month from the title silently misfiles rows.
- **The board's "% Increase/Decrease" column is sparse and inconsistent** (populated on a handful of JAWS rows). It is carried through for reference, but every percentage the report displays is computed from the turnover figures.
- **Months after the current one hold orders already booked, not turnover earned.** They are drawn outlined rather than filled and excluded from the year-on-year headline — summing them against a full prior year would understate the business badly. Only completed months (`lastCompleteMonthIndex`) feed the totals.

Entity is `JAWS` when the item name mentions JAWS, `VPS` otherwise; blanks stay `null` rather than collapsing to `$0`, so "no data" never reads as "no turnover". The year-on-year chart deliberately shows **two** series, not three: three years of one hue fail the normal-vision colour-separation floor ($\Delta E 14$ — indistinguishable even with full colour vision), so 2024 lives in the table beneath. Series colours are CSS variables with separately validated light and dark steps; note that CSS-var colours do not resolve in SVG presentation *attributes*, so every fill goes through `style={{{{}}`.

Where the reports get their data (audited 2026-08-20 when the workshop module was parked — nothing else needed reworking):

Source	Sections
MYOB (<code>lib/myob-reporting</code>)	KPI summary, P&L, top customers, receivables/payables aging, stock summary/reorder/dead, distributor ranking, pipeline, 6-month trends
Monday boards	Sales funnel, rep scorecards, quote aging, monthly conversion trend, combined pipeline
MechanicDesk facts (<code>md_invoices</code> / <code>md_quotes</code>)	Workshop Map, incl. Vehicle Trend
<code>calls</code> + <code>call_analysis</code>	All six Call Analytics sections
<code>crm_*</code>	Lead pipeline, campaign performance
<code>b2b_*</code>	B2B sales

None of them read the portal's workshop tables — that was only ever Workshop Performance, now removed.

7.9 Tasks & Projects (`/tasks` , `/projects` , `/todos`)

Tasks: Monday-style board/list + React Flow automations (5-min cron). Projects: force-directed web of the six Monday "Hidden To Do" boards with comment posting back to Monday. To-Dos: manager scorecards over the same boards.

7.10 Messaging (`/messages`) & Notifications

Portal-native channels/DMs (Slack replacement for internal chat; WhatsApp/Messenger/IG phases not built). Notification bell + badges with 8 emitters; Web Push for staff and distributors (PWA installable).

PWA link capturing (2026-08-25). Two manifests are served — `public/manifest.json` for staff (`scope: /`) and `public/manifest-b2b.json` for distributors (`scope: /b2b`), chosen per route in `_document.tsx` . Both now declare `"handle_links": "preferred"` so a Chromium browser opens in-scope URLs in the installed app instead of a tab, and `"launch_handler": { "client_mode": "navigate-existing" }` so a link reuses the open window rather than stacking new ones. Both also pin `"id"` to their existing `start_url` — the implicit id is `start_url` , so this changes nothing for an already-installed app while stopping a future `start_url` edit from re-identifying it as a different app.

Limits worth knowing before anyone reports it as broken: **iOS has no link capturing for home-screen web apps at all** — Safari always handles the link, and only a native wrapper would change that. On **Android** the WebAPK has to refresh before capturing starts (Chrome does this in the background, typically within a day; reinstalling forces it). On **desktop** the user may get a one-time "open in app?" prompt, and the behaviour is also togglable per-app in Chrome's app settings. Note the two scopes overlap — staff `/` contains `/b2b` — so on a device with *both* apps installed a `/b2b/...` link has two candidates; that combination shouldn't occur in practice (staff device vs distributor device) but it is the reason to be careful before widening the B2B scope.

7.11 Agents (`/agents`)

Monitoring-agent inbox (`lib/agent-framework` — not `lib/agents` , which is Pipeline A's mailbox map). Agents run every 15 min via cron; findings are triaged Dismiss / Mark done.

7.12 Stocktake — Mechanics Desk (`/stocktake`)

Upload count XLSX → Run Match (dispatches GH Action; includes coverage check) → review variance → Push to MD. Counted rows are sacred (row_number bug fixed 2026-07-30). MD is single-session — a stocktake run can 401 if someone logs into MD as the same employee mid-run (known, deferred).

7.13 Stripe→MYOB (`/stripe-myob`)

Lists Stripe invoices per account label and pushes them to MYOB as Professional Invoice + Customer Payment pairs; payout reconciliation endpoints support the JAWS accounts.

7.14 Sales/analysis surfaces

`/distributors` (group-aware revenue reporting; groups managed at `/admin/groups`), `/sales` , `/stock` (⚠ no SSR gate), `/forecasting` (MD job report forecast; `/jobs` redirects here), `/vehicle-sales` (platform classification cache from VPS invoices), `/job-reports` (MD job report ingest for PO→job matching).

`/distributors` grouping — one rule for every tab (2026-08-21). Groups live in `dist_groups` / `dist_group_members` (`/admin/groups`) across two dimensions: **type** = Distributors / Sundry / Excluded, **region** = National / International.

Until this change each tab had its own idea of scope, which is why the numbers never tied up:

Tab	Was
Summary	Sectioned by dimension, Sundry separate — correct
National P/M	Genuinely National-only, filtered server-side (<code>monthlyNational</code> in <code>pages/api/distributors.ts</code> keys off <code>location === 'National'</code> , excluding International and Sundry)
National Total	Titled "National" but contained everything , International and Sundry included

Tab	Was
Distributor Sales · Detailed Sales	Silently included Sundry
Parts : Tunes	Excluded Sundry via its own <code>typeOk</code> check

Now `sectionOfLine` in `pages/distributors.tsx` is the single definition, and a **Group by / Showing** bar under the tab strip drives every tab through `visibleLines`. Notes that matter:

- **Sundry is its own section in BOTH dimensions**, matching how the Summary has always rendered it. Otherwise a Sundry customer would be counted inside National or International and the sections would stop reconciling. Membership bears this out: all 23 Distributors carry a region, and the 15 customers with no region are all Sundry — so under region grouping there is no "Unclassified" section to explain away.
- **The monthly trend is now derived client-side** from the same `filtered` lines as everything else. The server's `monthlyNational` aggregate is hardcoded to National and could never follow the selection; it is still in the payload but deliberately unused. Fix the server aggregate too if anything else ever needs it.
- **Changing dimension resets the section to All** — a "Distributors" filter means nothing under `region`, and leaving it set would empty every tab.
- **Parts : Tunes keeps its Sundry exclusion while Showing is "All"** (Chris 2026-07-22), but honours an explicit section choice — otherwise picking Sundry would blank the tab and look broken.
- **Charts plot every customer; the totals are NUMBERS, not bars** (Chris 2026-08-21). `chartRows` carries both, flagged by `isTotal`, and every chart plots `chartRows.filter(r => !r.isTotal)` while `<ChartTotals/>` renders the group totals and grand total as a text strip beneath. Two failed attempts are worth not repeating: collapsing each group to a single bar destroyed the per-customer detail that was the whole point ("so you can see accurately what each made"), and total *bars* dwarf their own members — a group total is by definition larger than anything in it, so the axis rescales and every customer bar is squashed. In a pie a total slice is worse still: counted twice, percentages meaningless.
 - Category stacking (Tuning/Parts/Oil + custom) still applies within each customer's bar.
 - **A dashed divider separates one group's bars from the next** on both bar charts, drawn by `groupSeparatorPlugin()` — a hand-rolled Chart.js inline plugin, because the annotation plugin isn't loaded (Chart.js 4.4.1 comes from the CDN, so config-level `plugins: []` is the only hook). It reads the `section` field on `chartRows`, so a new grouping dimension needs no chart changes. Canvas can't resolve CSS variables, hence the literal stroke colour rather than a `T` token. Bars only — a pie has no axis to divide.
 - The **grand total row on the Summary table** is built from `distSummaries`, the same rows the sections above are built from, so it always reconciles to the sum of the section totals.
 - The **split mode of the National P/M trend is one line per GROUP**, not per customer — a 47-line time series is unreadable. It previously showed only the top 8 distributors, silently dropping the rest with no total to fall back on. Flag if per-customer lines are ever actually wanted.
 - The vehicle-model chart on Distributor Sales is a different dimension and is unchanged.
- **Negative categories are real and render as negatives.** CP Performance carries Parts −\$3,695.45 against Tuning \$9,454.55 (total \$5,759.10) from credits — the only negative in the payload as at 2026-08-21. A bar extending the other way there is correct, not a rendering fault (confirmed with Chris). Don't "fix" it by clamping to zero: that would hide a real credit and stop the chart reconciling with the table.

7.14a JA Assistant — removed from the UI (2026-08-21)

The floating AI assistant that sat bottom-right on every page is **no longer mounted**. It overlapped figures on the reports, so `<GlobalChatbot />` was taken out of `pages/_app.tsx`. Nothing else changed:

- `components/GlobalChatbot.tsx` still exists and still compiles. `ChatContextProvider` is still wrapped around the app, and the nine pages that call `useChatContext().setContext({...})` (calls, dashboard, distributors, overview, projects, reports, sales, stock, todos) are untouched — they set context that nothing now reads.
- **To bring it back, put `<GlobalChatbot />` back inside `<FeedbackProvider>` in `_app.tsx`.** That is the only change needed.
- Its API surface is now **unused but live**: `/api/chat.ts` and `/api/chat-sessions/*` (list, get, delete, send) plus their Supabase session/message tables. Left in place so the feature can be restored, and because the stored history is the user's. If the assistant is ever dropped for good, those routes, the tables and the nine `useChatContext` call sites are the cleanup.

7.15 Settings (/settings)

Tile launcher: General · Connections (Integrations / Health / MYOB) · Distributor Report config · VIN Codes · Users · Audit log · Profile · Claude connector (MCP tokens) · Service tokens · **Leave Notifications** (\$7.17) · **Library**.

7.16 Library (/admin/library)

This document and the SOP, served inside the portal — readable on screen with a contents rail, and downloadable as PDF. Registry: `lib/library-docs.ts` (one row per document). Gated on `admin:settings`.

- `docs/*.md` is the **source of truth**; the reader renders it live server-side (`marked`), so editing the markdown updates the on-screen copy at the next deploy.
- `docs/*.pdf` is a **generated artifact** — regenerate with `scripts/render-doc-pdf.js` (marked + Playwright; needs `npm install playwright chromium` once) and commit it. It goes stale silently otherwise.
- **⚠ The PDFs are deliberately not in `public/`.** This document names where every credential lives, the Supabase project id, the Tailscale addresses and the open security gaps; anything under `public/` is served to the internet to anyone holding the URL, with no sign-in. Downloads go through `/api/admin/library/[slug]`, which is auth-gated.
- **⚠ `next.config.js` → `outputFileTracingIncludes` must list `docs/**` for the three Library routes.** The paths are built at runtime so Next's tracer can't see them; without it the files are pruned from the serverless bundle and 404 in production while working perfectly in dev.
- **Illustrated documents.** Screenshots live in `docs/img/` and are referenced relatively (`![alt](img/foo.png)`). The renderer writes its print page to a temp file beside the markdown and loads it as a real `file://` URL — a `setContent()` page has an `about:blank` origin and silently renders file images blank. `DOC_DATE` overrides the cover's compiled date. Portal screenshots are produced from the real components (a throwaway page under `pages/` rendering the page component with mock props, driven by Playwright) — no live data is needed and none is captured.
- **⚠ The in-app reader does not rewrite relative image paths**, so an illustrated document renders its pictures in the PDF but shows broken images on screen. `docs/library-access-sop.md` (how to find these documents, with screenshots) is therefore **deliberately not registered** in `lib/library-docs.ts` — it is handed out as a PDF. Registering it needs an auth-gated `/api/admin/library/img/[file]` route plus a `renderer.image` rewrite in `pages/admin/library/[slug].tsx`.

SOP: see `CLAUDE.md` §1 — every change to the portal updates these documents as part of the same work.

7.17 Leave decision emails (Settings → Leave Notifications)

Staff apply for leave through the monday.com **Payroll & Leave Applications** board (`5027074711`) — a WorkForm drops the application into the *Leave Applications* group. When a manager sets the **Leave Approved** column to Approved or Denied, the portal emails the applicant. Built 2026-08-24 (migrations `197` , `198`).

- **Engine** `lib/leave-decision-emails.ts` , driven by `/api/cron/leave-decisions` every 15 minutes. Admin UI + manual run: `components/settings/LeaveNotificationsTab.tsx` → `/api/admin/leave-notifications` (`admin:settings`).
- **Why not a monday automation.** monday can only send mail through the Gmail/Outlook integration: it sends from one person's connected mailbox, dies quietly when that connection lapses, and does nothing at all when the board's Email Address column is empty — which is most rows, because managers hand-create them. The portal resolves the address itself instead.
- **⚠ What counts as an application.** That board doubles as a daily attendance log — "*Kaleb Rowe left work at 11am today*", "*Public Holiday*", "*Easter Monday - all staff*", "*TIME OFF/ OVERTIME EXPORT*" all sit on it marked **Approved**. Emailing on "status = Approved" alone would mail people about those. The rule (Chris's call) is *an item that was in Leave Applications and had Approved pressed on it*, and the board's own automations make it checkable: pressing Approved moves the item to **Upcoming Leave — Approved**, Denied moves it to **Leave Denied**. So only items in `topics` , `group_mkqz6qh6` or `group_mkqzjmed` (`APPLICATION_GROUPS`) are ever acted on; anything in the payroll groups is ignored outright — no email, no log row, no HR notice.
- **Address resolution:** the board's Email Address column → `leave_staff_directory` (name-as-typed → email, editable in the portal) → unresolved. A column address whose domain is a near-miss of ours is **rejected** rather than used (a live row reads `jarred@justautosmechanical.com.au` , which would bounce into a void with everyone believing the applicant was told). Directory matching is tiered: exact, trailing noise words stripped ("*Dom Simpson Sick*"), first + surname initial ("*Chris R*"), then a lone first name only when exactly one person has it — "*Matt*" (Huddy / Smith / Karger) stays unresolved on purpose, and a row naming several people ("*James, Kaleb, Graham, Dom and Tyronne*") refuses to match rather than mailing the first one. `scripts/check-leave-resolver.ts` exercises all of this against the real board names (`npx tsx` , no network).
- **Unresolved** items are logged `no_address` , HR is emailed **once**, and every later run retries — so adding the address to the column or the directory is all that's needed; nobody has to re-approve anything.
- **Dedupe / going live.** `leave_decision_emails` holds one row per (item, decision) and that is the dedupe key. On the very first run every already-decided application on the application path is written as `baseLine` and **nothing is sent** — 16 rows when this shipped. Flipping an item Approved → Denied is a new decision and does email again.
- Each send also posts an update on the monday item ("📧 Approval email sent to ..."), so the audit trail lives where HR is looking.
- **Settings** resolve DB-first through `integration_settings` : `LEAVE_EMAILS_ENABLED` (kill switch) and `LEAVE_HR_EMAIL` (copied on every email, the reply-to, and where unresolved notices go — `ryan@justautosmechanical.com.au`).
- **Known gap:** a separate board automation moves approved items into the payroll groups three days before the leave starts. If an approval were pressed and that mover ran inside the same 15-minute window, the item would leave the application path before the portal saw it and no email would go out. Practically impossible (a

daily automation vs a 15-minute cron) but real; the fix if it ever bites is a per-item Send button on the Leave Notifications screen.

7.18 Stock EOM — JAWS month end (/reports/jaws-stock-eom)

Month-end stock report for the **JAWS** company file. Built 2026-08-24 (migration 199). Engine `lib/jaws-stock-eom.ts` ; API `/api/reports/jaws-stock-eom` ; cron `/api/cron/jaws-stock-eom` ; **PDF export** `/api/reports/jaws-stock-eom/pdf?month=YYYY-MM` rendering `lib/jaws-stock-eom-pdf.tsx` (`@react-pdf/renderer` , house style shared with `lib/reports/pdf.tsx`).

Why it exists alongside /stock . `/stock` (`pages/api/inventory.ts`) already computes the *live* picture — reorder alerts, velocity, dead stock, margin, on-order — on rolling 30/90/365-day windows. What it cannot do is compare months, because **AccountRight only ever reports today's quantity**: there is no historical on-hand to query. So each run freezes its numbers into `jaws_stock_snapshots` , and that stored history is the only source of month-on-month stock movement in the business.

What the report adds beyond the live page: the month's trading in isolation (units, revenue ex-GST, COGS, margin); stock turn and days-of-inventory; ageing of held value by last-sold date; slow movers ranked by capital at risk; margin leakage (sold below cost) and cost creep (last paid >10% above average cost — a price-review list); unfilled demand (sold while nothing available, or committed beyond on-hand); overstock (>365 days cover); supplier concentration of value and reorder spend; data-integrity exceptions (negative on-hand, stock with no cost, stock with no sell price); and any JAWS stocktake completed that month (migration 141 , still report-only).

Reuse, deliberately. It calls the same `fetchInventoryItems` / `fetchSaleInvoicesWithLines` readers and the same `lineExGst` normalisation as `/stock` , so the two surfaces reconcile instead of becoming a second, subtly different truth.

Sales-history window (Chris, 2026-08-25). Every average, the months-of-cover figure and the growth read are measured over a window chosen on the report — `?from=YYYY-MM&to=YYYY-MM` , presets 3/6/12/24 months, or two month boxes. Defaults to the 12 months ending with the reported month; `MAX_HISTORY_MONTHS` caps it at 36 (trimmed from the front, keeping the end fixed) so the MYOB read stays inside the 300s budget.

`resolveHistoryWindow()` clamps `to` to the reported month — a month-end report must never average in sales it could not have known about — and is exported so the API can compare the requested window against a stored snapshot's before deciding to reuse it. **A window the snapshot wasn't built with forces a rebuild**, on the screen and on the PDF export alike; otherwise the page would silently show figures for a different period.

Per SKU: `historyUnits` / `historyRevenueEx` , `avgUnitsPerMonth` / `avgRevenuePerMonth` (÷ months of the *window*, so a month with no sale counts as a zero), `monthsCoverAtAvg` (on-hand at the average rate — steadier than `daysOfCover` , which is the last 90 days alone; both are shown, and the difference is the seasonality signal), and `growthPct` . Report-level: a `history` block with the month-by-month `series` behind the whole-business growth read, surfaced as a *Sales by month* table on screen, in the PDF and summarised in the email.

`halfOverHalfGrowth()` compares the back half of the window with the front half, dropping the middle month on an odd count so the halves are equal, and returns null under 4 months — two-month halves are noise. The 13-month invoice read still happens regardless of the window, so stock turn stays comparable; a longer window simply starts the read earlier. Window, average and growth are stored per snapshot (migration 202) — without them an old snapshot's averages can't be interpreted.

Stock position — the over/under-stocked read (Morgan's ask via Chris, 2026-08-25). Every `EomItem` now carries `monthlySeries` (units + ex-GST revenue for every month of the window), and `stockPosition(item, n)` reads the last `POSITION_MONTHS` (6) of it against on-hand: over 6 months of cover is `Overstocked` , under 1

month `Short` , no sales in the period with stock held is `No sales` . `stockPositionList` on the report ranks Short → Overstocked → No sales → OK, then by capital, capped at `LIST_CAP` . It renders as a month-by-month grid in **the month-end email** (which is where it was asked for — Morgan gets the six columns, on-hand, average and verdict without opening the portal), and on the report screen and the PDF. It answers a different question from the slow-mover list: that ranks capital at risk over the whole window, this asks whether the holding matches recent demand. The email also prints the last 6 months of the whole company file's sales with the month-on-month change.

Dropped from display, still computed (Chris, 2026-08-25): *Value and spend by supplier* and *Data to fix in MYOB* are gone from the report screen and the PDF — neither drove a decision at month end. `suppliers` and `integrity` are still built and stored on the snapshot, so reinstating either is a render away with no rebuild.

On-hand quantity now appears in most tables (same change) — top movers, margin earners, slow movers, below cost, cost creep, unfilled demand — labelled "as at" the generation date, since AccountRight only ever reports today's quantity.

PDF export (2026-08-25). *Export PDF* on the report page downloads the whole report — headline figures, ageing, every exception list and the notes — as A4. It serves the **stored snapshot**, so the PDF always matches what is on screen and returns in about a second; it only builds live when that month has never been generated. The renderer is tolerant of pre- 201 snapshots (missing `capitalAtRisk` / `slowCapital` / `analysedValue` print as `-` rather than `NaN`), so an old month exports rather than failing. The button fetches with credentials and downloads a blob rather than linking directly, so a permission or build error surfaces as a toast instead of a browser error page.

Never-sold stock is excluded from the "not moving" analysis (Chris, 2026-08-25). A SKU on this item list that has never been invoiced is almost always a kit component never sold separately — `P-INTK-PIPE` (intake pipe *only*), `17276-52010` / `96711-35053` (intake gaskets No.1 and No.2), the distributor cutting jigs — and it dominated the dead-stock figure while carrying no possible action: **\$47.6k of July's \$114.7k**. Ageing, dead-90, dead-180, overstock and slow movers now run over `soldEver` (held stock with a last-sold date), and ageing shares are of `analysedValue` rather than the whole holding, so the buckets still total 100%. The excluded count and value stay in the headline (`neverSoldCount` / `neverSoldValue`) and in the notes — the capital is reported, never silently dropped.

Slow movers are a capital measure, not a silence measure (same change). The old list was simply "nothing sold in 90 days", which missed the biggest problem in the business: SKUs that sell *steadily* while carrying a year of stock. Each item now has `capitalAtRisk` = value held beyond `TARGET_COVER_DAYS` (90) of its own demand, capped at MYOB's `CurrentValue` ; because `runRatePerDay` is 0 for a dead SKU this equals the whole stock value with no special case. A SKU joins the list when nothing sold in the 90 days to month end **or** it holds over `SLOW_COVER_DAYS` (180) of cover with at least `SLOW_CAPITAL_MIN` (\$2,000) past the target, and the list ranks by capital at risk. Against July's real numbers the top three — FJA300 Intake Pipe (\$67.0k at risk, 378 days cover), 4" DPF Revision (\$66.7k, 371 days), F33A Sump Kit (\$59.8k, 268 days) — were all **invisible to the old rule**, because all three were selling. Overstock (>365 days) is the extreme end of the same list and is still shown separately. `slow_count` / `slow_capital` land in the snapshot (migration 201).

Comparing months across the change. Snapshots written before 2026-08-25 hold the old semantics — `dead_90_value` included never-sold stock — so the trend line and the month-on-month delta step down at the change point. Rebuild an old month from MYOB (Reports → Stock EOM → **Rebuild from MYOB**) to restate it on the new basis.

Reorder scope. Suggestions are drawn **only from the Stock Order sheet** (`b2b_reorder_items` , migration 114) — the curated list of SKUs the business actually buys. MYOB's item list is far wider and includes **kit components that are never sold separately**; those sit below their alert level permanently and swamped the list on first use (Chris, 2026-08-24). `b2b_product_bundles` cannot be used to identify them — it holds one row. Off-sheet items below their alert level are counted (`reorderExcludedCount`) and the count is shown, so a SKU that *should* be ordered is still visible as a number; the fix is to add it to the Stock Order sheet. If the sheet is ever empty the report falls back to every item and says so in its notes, rather than silently reporting "nothing to buy".

⚠ **Sales figures are bounded to the reported month.** `lastSold` , the 90/365-day windows and days-since-last-sold all stop at month end, so re-running an old month gives the same answer. This was a real bug on first use: the MYOB fetch runs to *today*, so an item sold after the month closed produced a last-sold date beyond the month end and a **negative** "days since last sold" (–11 on a July report). Each item also carries `unitsSinceMonthEnd` , surfaced as a **Sold since** column on the slow-mover and never-sold tables — a slow mover that has started selling again is then obvious instead of looking dead.

⚠ **Two approximations, printed on the report itself:**

- **On-hand is "as at generation time", not the last instant of the month.** Unavoidable — see above. The cron runs early on the 2nd to keep the gap small.
- **COGS = units × current average cost.** Invoice lines carry no cost of sale and average cost drifts, so margin ranks SKUs reliably but is *not* the P&L. Don't reconcile it to the accounts.

Access. Page and API both require `view:stock` , not just `view:reports` — the report carries costs, margins and supplier pricing, so a reports-only login (e.g. marketing) is refused. The tab is admin/manager only, and it participates in the per-user report allowlist (`visible_report_tabs`).

Settings (DB-first via `integration_settings`): `JAWS_EOM_EMAIL_TO` (comma-separated; defaults to Chris + Morgan), `PORTAL_BASE_URL` for the email's deep link. A month can be rebuilt on demand from the page ("Rebuild from MYOB") or re-run for a specific month via `?month=YYYY-MM` on the cron route.

8. Monitoring & troubleshooting

1. **First step:** `/admin/connections` (or Settings → Connections → Health). 21 checks across accounting / workshop / comms / crm / phone / infra, written every 5 min by the health cron. Manual force: `curl -H "Authorization: Bearer $CRON_SECRET" https://justautos.app/api/cron/health-check?force=1`.
2. **Freshness-based checks** (PBX CDR sync, transcribe, Deepgram, MD pulls) go red when data stops arriving — the fix is on the source host/worker, not the portal.
3. **Graph mailbox rows red** → renewal cron failed or the subscription died: re-run `setup-graph-subscriptions` (idempotent).
4. **MYOB rows red** → token lapsed: re-run the connect flow (Settings → MYOB Connection). Remember refresh tokens die if unused for weeks.
5. **GH Actions rows red** → check the Actions tab on the repo; MD workers upload failure artifacts and Slack on failure.
6. **A B2B order paid but no MYOB invoice** → `b2b_orders.myob_write_error` ; fix cause, retry from the admin order page. Stripe webhook itself is idempotent.
7. **Prints not coming out** → the workshop PC agent: check the tray/NSSM service; it drains pending jobs on startup, so restarting it usually clears the backlog.

9. Known risks, debt & outstanding items (as of 2026-08-24)

Security

- Supabase `service_role` key was exposed in an April 2026 session and **has never been rotated**. Rotation must update: Vercel env, the FreePBX host workers, the workshop print agent `.env`, and GitHub Actions secrets — all hold it.
- Repo working tree contains `agents/label-print-agent/.env` and `hardware/ja-scale-node/secrets.h` — verify both are gitignored and never committed with live values.
- `NEXT_PUBLIC_GOOGLE_PLACES_API_KEY` is client-exposed — must stay referrer-restricted.
- Three pages lack SSR auth gates (`/stock`, `/imports`, `/admin/connections`) — API-level gating covers data but the pages render.

Consistency / drift

- `lib/auth.ts` role type is missing `workshop` (use `lib/permissions.ts`).
- Migrations `148` and `153` are duplicated numbers; latest applied is `202_jaws_stock_eom_history_window`, so next is `203`.
- **Other selects still relying on a big `.limit()`** rather than `selectAllRows()`. Paged 2026-08-24: the weekly quotes/jobs map report and the weekly calls report (both were already truncating), and the overnight-leads store (`lib/sales-recap-leads-store.ts` — its read is unbounded, so a 3-month custom range in Reports → Sales Report was heading for ~2.6k rows). Remaining sites are safe only while their tables stay small — measured 2026-08-24, all well under the cap: `lib/crm-campaigns.ts` and `lib/reports/fetchers.ts` (×2) plus `pages/api/crm/campaigns/index.ts` (CRM tables still empty), `pages/api/imports/[id]/{run,finalize}.ts` (max 78 chunks per import), `pages/api/notifications/summary.ts`, `pages/api/b2b/admin/stock-transfer.ts` (109 rows), `pages/api/workshop/purchase-orders/generate-low-stock.ts` (0 rows). Re-check before any of them grows.
- `bank-payments-slack` and `slack-cleanup` cron handlers exist but aren't scheduled in `vercel.json`.
- Seven overlapping base-URL env vars.
- The Library reader can't display images (no rewrite of relative `img/` paths, no auth-gated image route), so `docs/library-access-sop.md` sits in `docs/` as a PDF-only document rather than on the Library shelf — see §7.16.
- Older docs (`03`, `05`, `07`) predate the CData decommission, B2B rollout and Xero foundation, and still describe the workshop migration as active — trust this document and the code.

Operational

- MYOB and Xero OAuth tokens need periodic exercise; Xero refresh tokens are single-use (rotation is handled, but never hand-edit the row).
- The FreePBX box is CentOS 7 — camera bridges pinned to Node 16; the whole host is a single point of failure for calls, coaching, and camera alerts. `sip-loss-monitor.sh` may still be running there from the phone-dropout investigation (upstream/FreedomBroadband was the cause) — clean up when resolved.
- **Queued-then-parked calls report their longest single leg, not the whole conversation.** `classifyCall` picks the longest-billsec `Dial / Park` leg with a `dstchannel`; on a queue call the leg carrying the true total has `lastapp='Queue'` and is excluded. Two known cases: 2026-07-20 09:30 shows 14:33 on Kaleb against a

real ~30:19, and 2026-08-10 14:50 shows 25:16 on Tyronne against ~32:55. Both duration *and* advisor attribution are affected, so correcting it would move historical coaching numbers — deliberately left alone.

- **MechanicDesk is staying** (decision 2026-08-20) — the replacement build is paused, so the 9 scheduled MD scrapers are a permanent production dependency rather than a temporary bridge. The portal's workshop module is built but unused; `docs/workshop_md_parity.md` keeps the cutover checklist if it is ever revived.
- Parked/off: negative-call automation (`CALL_CONCERNS_ENABLED`), portal-side calls analysis cron, Places key on New Booking quick-add (key unset), first real JAWS→VPS stock transfer pending, Live Bins hardware untested in production, live call monitoring WSS keep-alive retest pending.
- MYOB→Xero migration: foundation only — adapter waves per module still to come; both entities eventually move.

10. Key file index

Area	Files
Auth/permissions	<code>lib/permissions.ts</code> , <code>lib/authServer.ts</code> , <code>lib/auth.ts</code> , <code>lib/b2bAuthServer.ts</code> , <code>lib/b2bSupplierAuth.ts</code> , <code>lib/service-auth.ts</code>
Credential resolver	<code>lib/integration-config.ts</code> , <code>pages/api/admin/integrations.ts</code> , <code>components/settings/IntegrationsTab.tsx</code>
Accounting	<code>lib/myob.ts</code> , <code>lib/myob-reporting.ts</code> , <code>lib/xero.ts</code> , <code>lib/accounting-provider.ts</code> , <code>lib/accounting/*</code>
B2B pipeline	<code>lib/b2b-order-pipeline.ts</code> , <code>lib/b2b-payment.ts</code> , <code>lib/b2b-myob-invoice.ts</code> , <code>lib/b2b-machship.ts</code> , <code>lib/b2b-freight-*.ts</code> , <code>lib/b2b-dropship-*.ts</code> , <code>lib/b2b-tune-jobs.ts</code>
AP	<code>lib/ap-extraction.ts</code> , <code>lib/ap-auto-entry.ts</code> , <code>lib/ap-statement-watch.ts</code> , <code>lib/ap-myob-bill.ts</code>
Workshop tooling	<code>pages/workshop/letters</code> , <code>.../prepick</code> , <code>.../purchase-orders</code> , <code>.../stocktake</code> , <code>lib/workshop-*.ts</code> . Paused replacement build: rest of <code>pages/workshop/*</code> + <code>docs/workshop_md_parity.md</code>
Calls	<code>lib/live-calls.ts</code> , <code>lib/calls-analysis.ts</code> , <code>lib/softphone.ts</code> , <code>docs/pbx_click_to_dial_worker.md</code>
Mail	<code>lib/microsoft-graph.ts</code> , <code>lib/email.ts</code> , <code>lib/agents.ts</code> (mailbox→owner map)
Health	<code>pages/api/cron/health-check.ts</code> , <code>pages/admin/connections.tsx</code>
Workers	<code>.github/workflows/*</code> , <code>scripts/*</code> , <code>agents/label-print-agent/</code> , <code>agents/ja-freightbay/</code> , <code>hardware/ja-scale-node/</code>
Config	<code>vercel.json</code> (crons + maxDurations), <code>migrations/</code>